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A reliable cyclic envelope model for partially-restrained steel beam-column bolted T-stub connections based on experimental data

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ABSTRACT

The utilization of partially-restrained (PR) steel beam-to-column bolted T-stub connections offers a promising avenue for enhancing the efficiency and resilience of steel structural systems by reducing material consumption and facilitating post-hazard rehabilitation of connection components. However, the practical application of PR T-stub connections faces challenges due to the lack of comprehensive design guidelines. Specifically, in the context of performance-based seismic design for structural systems featuring PR T-stub connections, the accurate characterization of the cyclic envelope is pivotal. Presently, the availability of dependable cyclic envelope models tailored to PR T-stub connections is limited, with existing models facing various limitations. In response to these challenges, this study aims to pioneer a more dependable envelope model capable of mitigating the shortcomings of existing models. To achieve this objective, a comprehensive database comprising experimental data from steel beam-to-column assemblages with T-stub connections subjected to cyclic loading is compiled. These assemblages feature T-stubs either cut from hot-rolled sections or fabricated by welding plates, while the connected beams and columns are hot-rolled members. A four-stage multi-linear cyclic envelope is then developed. Particular attention is given to determining the two key characteristics of PR T-stub connections: maximum strength and initial rotational stiffness. The accuracy and robustness of the developed envelope model are rigorously assessed through comparisons with both experimental results and predictions from existing envelope models.

1. Introduction

Among a variety of partially-restrained (PR) steel beam-column connections, also referred to as semi-rigid connections, the bolted T-stub connection (illustrated in Fig. 1) is one of the favorable connections partly because field welding is eliminated. Specifically, T-stub components are bolted to both the column flange and the beam flanges using bolts, and the beam web is usually bolted to the column flange through a shear tab that is welded to the column in the shop. The use of such T-stub connections facilitates prefabricated steel constructions with rapid construction speed. Moreover, it also makes the deconstruction easier compared to welded connections, thus rendering convenient post-hazard rehabilitation.

Based on the strength, a PR T-stub connection is further classified as PR full-strength if its strength exceeds the fully plastic moment strength of the connected beam; otherwise, it is considered PR partial-strength. Frames with PR partial-strength T-stub connections may possess higher damping [1], as T-stub components with finite strength can serve as a reliable source of energy dissipation under cyclic loading during

earthquakes [2–3]. In addition, frames incorporating PR T-stub connections also attract lower seismic forces due to the period elongation caused by the flexibility in PR connections, compared to rigid frames. Furthermore, the potential for favorable bending moment distributions offered by PR T-stub connections may result in economical cross-sectional size for steel members [4]. Both the structural and economic performance potentials have spurred experimental and analytical studies of PR T-stub connections at the system level (e.g., [5,6]) or at the sub-assembly level (e.g., [7–9]).

Despite the above-mentioned advantages, the practical implementation of PR T-stub connections faces challenges due to their complex behavior and the lack of comprehensive design guidance. Particularly, a main challenge is the development of reliable cyclic moment-rotation (force-displacement) envelope models for PR T-stub connections. In the context of the notable shift from conventional force-based seismic design towards performance-based seismic design (PBSD) [10–13], nonlinear dynamic analysis is essential for performance assessment and verification [14]. In nonlinear dynamic analysis, computational efficiency is crucial for performance-based design-oriented analysis models. As such, PR beam-column connections in

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Notation			
A_b	Gross area of the bolt shank	n_{sb}	Total number of shear bolts on the T-stem
$A_{b,eff}$	Effective area of the bolt shank	n_{tb}	Total number of bolts on the T-flange
A_n	Net section area of the T-stem	r	Radius of the K-zone of the T-stub
$A_{n,eff}$	Effective net section area of the T-stem	t_f	Thickness of the flange (t_{Tf} for the T-stub, t_{bf} for the beam, and t_{cf} for the column)
c_{be}	Coefficient accounting for strain-hardening benefits for the beam	T_i	Force applied to the T-stubs ($i = 1, 2, \dots, 6$)
c_{fl}	Coefficient accounting for strain-hardening benefits for the T-flange	T_i'	Force applied to Each segment of the T-flange ($i = 1, 2, 3$)
c_r	Residual strength parameter of the envelope	t_s	Thickness of the T-stem
D	Cross-section depth (D_T for the T-stub, D_b for the beam, and D_c for the column)	t_w	Thickness of the web (t_{bw} for the beam and t_{cw} for the column)
d_{tb}	Diameter of the tension bolt shank	w	Length for a single divided segment of a T-flange
d_{hole}	Bolt hole diameter	W	Cross-section width (W_T for the T-stub, W_b for the beam, and W_c for the column)
FM1	T-flange forming plastic mechanism	Z	Distance between the center of compression and tension
FM2	T-flange yielding governed by tension bolt fracture due to prying force	Z_p	Plastic section modulus about major axis
FM3	Tension bolt fracture without prying force	α	Hardening parameter of the envelope
FM4	T-stem net section fracture	β	Post-peak softening parameter of the envelope
FM5	Shear bolt fracture on the T-stem	δ	Coefficient considering material loss
FM6	Formation of a plastic hinge at the beam	θ	Connection rotation
f_y	Yield strength of the T-stub and member (beam and column)	θ_c	Rotation corresponding to M_{max}
f_u	Ultimate tensile strength of the T-stub and member (beam and column)	θ_r	Rotation corresponding to $c_r M_{max}$
f_{ub}	Ultimate tensile strength of the bolt	θ_u	Ultimate rotation
f_{vb}	Nominal shear strength of the bolt	θ_y	Rotation corresponding to M_{y-equi} (rotation at the endpoint of the initial elastic stage of the envelope if M_{y-equi} is not available)
k_i	Stiffness coefficient of the component (the subscript of k corresponding to that of Δ)	$\Delta_{cw,c}$	Transverse compression deformation of the column web
K_{ini}	Initial rotational stiffness	$\Delta_{cw,t}$	Transverse tension deformation of the column web
m	Distance from a point in the K-zone to the edge of the bolt or to the centerline of the bolt hole	Δ_{cf}	Column flange bending deformation due to tension force,
m_0	Distance from the T-stem face to the centerline of the bolt hole	Δ_{Tf}	T-flange bending deformation
M_{max}	Maximum moment	$\Delta_{bol,s}$	Deformation of the bolt in shear
M_p	Cross-sectional full plastic bending moment	Δ_{Ts}	Deformation of the T-stem in tension
M_{y-equi}	Equivalent yielding moment	$\Delta_{bol,t}$	Deformation of the bolt in tension
n	Distance from the T-flange edge to the edge of the bolt or to the centerline of the bolt hole	$\Delta_{Ts,be}$	T-stem hole deformation due to bearing around bolts
n_0	Distance from the T-flange edge to the centerline of the bolt	$\Delta_{bf,be}$	Beam flange hole deformation due to bearing around bolts
		Δ_{slip}	Slip between the T-stem and beam flange
		$\sum \Delta_{Ten}$	Sum of deformations of the connection components in tension
		$\sum \Delta_{Comp}$	Sum of deformations of the connection components in compression

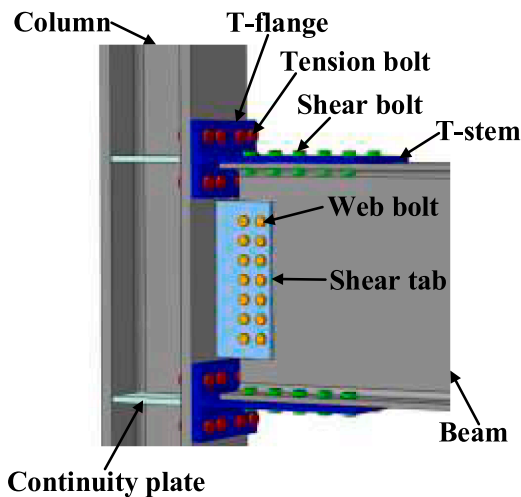


Fig. 1. Schematic view of a steel beam-column bolted T-stub connection.

frames are commonly represented by multiple springs [15], which often rely on the cyclic envelope of the PR connections integrated with appropriate hysteretic rules [2,16–18]. Therefore, the cyclic envelope of PR connections plays a vital role in assessing the seismic performance of steel frames with PR connections and its accurate characterization is essential for the successful implementation of these structural systems in practice.

Currently, there is limited cyclic envelope models tailored for PR T-stub connections. ASCE 41-17 [13] introduced a four-stage envelope model for PR T-stub connections, following the guidelines specified in FEM 356 [19]. Leon [20] provided two distinct envelope models for T-stub connections, developed based on experimental studies conducted by Smallidge [21] and Swanson [22]. However, these models have several limitations when it comes to determining the critical characteristics of PR T-stub connections, such as the maximum moment (M_{max}) and the initial rotational stiffness (K_{ini}), which are of utmost concern to engineers when assessing the performance of these connections.

In both [13] and [20], M_{max} for PR partial-strength T-stub connections is determined without accounting for either the material strain-hardening benefits or the material loss due to bolt holes in the T-flange, postulating that their effect is negligible for simplicity.

Although it can be argued that the strength enhancement from strain-hardening compensates for neglecting the material loss from holes, this assumption should undergo further validation. Additionally, M_{max} for PR full-strength T-stub connections may also be significantly influenced by the steel strain-hardening of the connected beam. Leon [20] adopted an amplification factor of 1.1 for determining M_{max} for full-strength T-stub connections, while ASCE 41-17 [13] employed amplification factors ranging from 1.12 to 1.27. The suitability of these amplification factors for a broad range of PR full-strength T-stub connections requires further evaluation.

Similarly, different criteria have been proposed in the literature regarding K_{ini} for T-stub connections. In ASCE 41-17 [13], K_{ini} for PR partial-strength T-stub connections is determined based on M_{max} , assuming an empirical value of 0.5% for the corresponding rotation, while K_{ini} for PR full-strength T-stub connections in [13] is determined as six times the bending stiffness of the beam. In contrast, K_{ini} for partial-strength T-stub connections provided by Leon [20] is determined using the formula $0.8M_{max}/(7.62/D_b)$, where D_b represents the beam depth in millimeters, while K_{ini} in [20] for a full-strength T-stub connection is determined as $0.65M_{max}/(0.015)$, assuming an empirical value of 1.5% for the rotation at $0.65M_{max}$. The reliability of these empirical values needs to be further assessed.

In addition to [13] and [20], several other methods exist for determining the two critical characteristics, M_{max} and K_{ini} , of the PR T-stub connections. For example, EN 1993-1-8:2005 (E) [23] was employed to calculate the maximum strength of PR T-stub connections subjected to cyclic loading in some studies (e.g., [24]). However, it should be pointed out that EN 1993-1-8:2005 (E) [23] is theoretically not applicable to determining cyclic M_{max} . This is because it was mainly developed based on monotonic test results. Note that EN 1993-1-8:2005 (E) [23] recommends the use of the component method [25–26] to determine K_{ini} for a connection. This method is based on the idea that a connection's flexibility can be broken down into contributions from its individual components. One notable advantage of the component method is its ability to consider pertinent constituent component of a specific type of connection, potentially providing more reasonable prediction of K_{ini} . However, the component method-based K_{ini} provided in [23] has some limitations, as T-stem elongation in tension and the slip between T-stem and beam flange are not considered. Although the stiffness model provided in [22] accounts for the deformation of these two components, it does not incorporate the influence of column flange deformation. Different to the component method, Wang [27] attempted to predict K_{ini} for PR T-stub connections using a simplified equation, but this equation was developed based on limited experimental data and led to poor accuracy as revealed later in this study.

In order to verify the accuracy of existing envelope models, particularly the two critical characteristics (M_{max} and K_{ini}) of the envelope, experimental tests typically serve as the point of departure. The experimental data compiled can also be utilized to develop a more reliable envelope model to address the deficiencies of existing methods, if any. A comprehensive review of the literature reveals although some researchers have established experimental databases for different types of PR connections, there is a lack of experimental data compiled specifically for T-stub connections. For instance, Nethercot [28] summarized over 700 riveted and bolted PR connection tests conducted prior to 1984, which included a few T-stub connections comprising older grade bolts subjected to monotonic loading. Chen and Kishi [29] compiled a database of 303 PR connections with seven different types of connections tested between 1936 and 1989. Abdalla and Chen [30] later expanded this database. The majority of the collected specimens in the expanded database were subjected to monotonic loading, and none of the tests collected were related to T-stub connections. Weynand et al. [31] developed a databank system named SERICON II, which includes data sets of PR connections prior to 1998, but it lacks details about the PR connections. More recently, Mak and Elkady [32] created a comprehensive experimental database for PR connections called

SRConED, with a focus on flush end-plate connections tested between 1970 and 2020. This database includes 64 specimens that were subjected to cyclic loading. In addition to these databases that mainly focus on PR connections, the beam-to-column connection database created by the SAC Steel Project [33] and the database established by Lignos and Krawinkler [34] primarily focused on fully restrained (FR) welded connections, with only a few test data sets for PR bolted connections. However, no T-stub connections were covered in the two databases.

With the specific goal to develop a reliable cyclic envelope model for PR T-stub connections, the present study compiles an experimental database of PR T-stub connections subjected to cyclic loading. The reliable cyclic envelope model is aimed at addressing the deficiencies found in existing envelope models for PR T-stub connections. As such, the remainder of this work is organized as follows. Section 2 summarizes the experimental data obtained from a comprehensive series of tests on PR steel beam-to-column assemblages with T-stub connections subjected to cyclic loading. Subsequently, the developed envelope model and other existing models are briefly described in Section 3. The proposed method together with other methods for determining the two prominent characteristic properties (M_{max} and K_{ini}) and other parameters of the envelope model are presented in Section 4. The prediction of M_{max} and K_{ini} by different methods are assessed through their comparison against a wide range of experimental results. Before concluding in Section 6, the envelope model developed is validated against experimental results. Its accuracy and robustness are also demonstrated by comparing it with other envelope models. The results of this research can be further used to develop cyclic hysteretic models for PR T-stub connections in the future together with classical or newly proposed hysterical rules, which will facilitate seismic performance assessment of steel frames incorporating PR T-stub connections, eventually contributing to the design methods for PR steel frames.

2. Experimental database compiled for PR T-stub connections subjected to cyclic loading

The database compiled contains a total of 47 tested connections from 11 different cyclic experimental studies carried out from 1994 to 2019. The test configurations and the number of connections reported in each experimental study are summarized in Table 1. The types of test configurations are represented as DS, SS1, SS2 and SS3, respectively, as illustrated in Fig. 2. Note that a double-sided beam-to-column assemblage, with two beams connected to a column on each flange, is listed as having two connections in the database. Further details of the collected specimens are shown in Appendix A. All the collected T-stub connections are classified as partially-restrained (semi-rigid) connections. It should be pointed out that four connections with T-stubs connected to the column web reported in [35] are excluded from the database because they are beyond the scope of the present study.

Table 1

Summary of the test configurations for the PR T-stub connections compiled.

Ref.	Test configurations	No. of connections
[36]	DS ⁽¹⁾	4
[21]	SS2	6
[37–38]	SS2	2
[24]	SS3	2
[9]	SS1 ⁽²⁾	1
[39]	SS3	2
[40]	SS1	6
[41]	SS1	7
[42–43]	DS	4
[35]	DS ⁽³⁾	12
[44]	DS ⁽³⁾	4

Note: ⁽¹⁾: Column top end was pinned and cyclic load was applied at the column bottom end ⁽²⁾: Column was placed on the floor; and ⁽³⁾: Cyclic loads were applied in turn at the major axis plane and the minor axis plane of spatial assemblages.

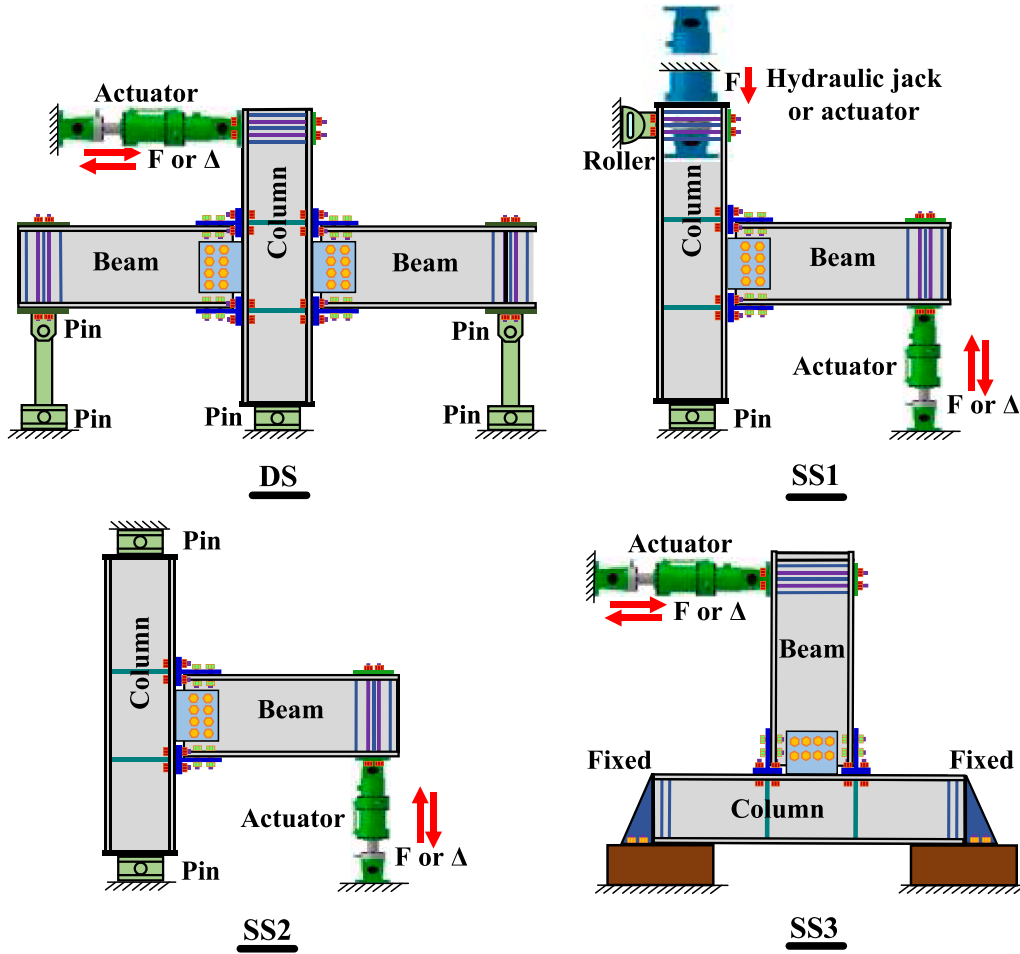


Fig. 2. Types of test configurations for the collected T-stub connections (the details of the actual connections, e.g., the number of the bolts and steel stiffeners, varied for different tests).

2.1. Brief summary of the tested specimens compiled

In the database compiled, all beams and columns connected to the T-stub connections were made from hot-rolled H-shaped sections, also referred to as “W” sections in AISC [45]. Hot-rolled sections were predominantly used for the T-stubs as well, with the majority being cut from H-shaped sections, as detailed in Table A1 of Appendix A. Additionally, a few T-stubs were constructed by welding plates together. It is important to note that the welds in these built-up T-stubs remained intact throughout the testing process, with no weld failures reported. The geometric dimensions of the collected specimens have a good coverage of steel members or connections, as shown in Table A2 and A3 of Appendix A. Specifically, the column cross-sectional dimensions, D_c , W_c , t_{cw} and t_{cf} ranged from 200–800 mm, 200–450 mm, 9–35 mm, and 12–63 mm, respectively. Here, the symbols D , W , and t denote cross-section depth, cross-section width, and web (or flange) thickness, respectively. For beams, D_b , W_b , t_{bw} and t_{bf} varied between 269 and 911 mm, 132–305 mm, 7–16 mm, and 10–24 mm, respectively. As for T-stubs, D_T , W_T , t_{Ts} and t_{Tf} were 150–730 mm, 178–340 mm, 8–24 mm, and 12–50 mm, respectively. The structural steel yield strength (f_y) and ultimate tensile strength (f_u) ranged from 260 to 441 MPa, and 410 to 565 MPa, respectively. Regarding the bolts, high-strength bolts with a grade of 10.9 in Europe [23] and China [46] or A490 in North America [47,48] were used in all the tested specimens, except the two specimens reported in [36] employed A307 and A325 bolts. The grade and size of the bolts used in each specimen were consistent, with the range of the bolt shank diameter being 19.1 mm to 35 mm. All the bolts were applied by

pretension force or torque.

2.2. Experimental moment and rotation

The moment resisted or transferred by the connection is defined as the column flange face moment, while the connection rotation (θ) is defined as the relative rotation at the beam-column interface, as illustrated in Fig. 3. θ is caused by deformations from different components that contribute to the connection rotation. The component deformations considered for θ include T-flange flexure, column flange flexure, transverse compression of column web (in case of no web stiffener), transverse tension of column web (in case of no web stiffener), T-stem elongation, tension bolt elongation, shear bolt deformation, slip between the T-stub and beam flange, and deformation of the T-stem holes and beam flange holes due to bearing around bolts.

Of the test specimens, 29 reached their maximum moment resistance, while the remaining 17 did not undergo failure during cyclic loading either because the predetermined loading protocol was finished or due to the capacity limitation of the test machine. The experimental maximum moments ($M_{max-Exp}$) for these two groups are distinguished in the database. More information regarding $M_{max-Exp}$ and its corresponding connection rotation (θ_{c-Exp}) is provided in Table A3 of Appendix A. It should be noted that the values of $M_{max-Exp}$ and θ_{c-Exp} provided in this table are the averages of both positive and negative directions.

3. Description of the envelope models

Fig. 4 shows the cyclic envelope models for T-stub connections,

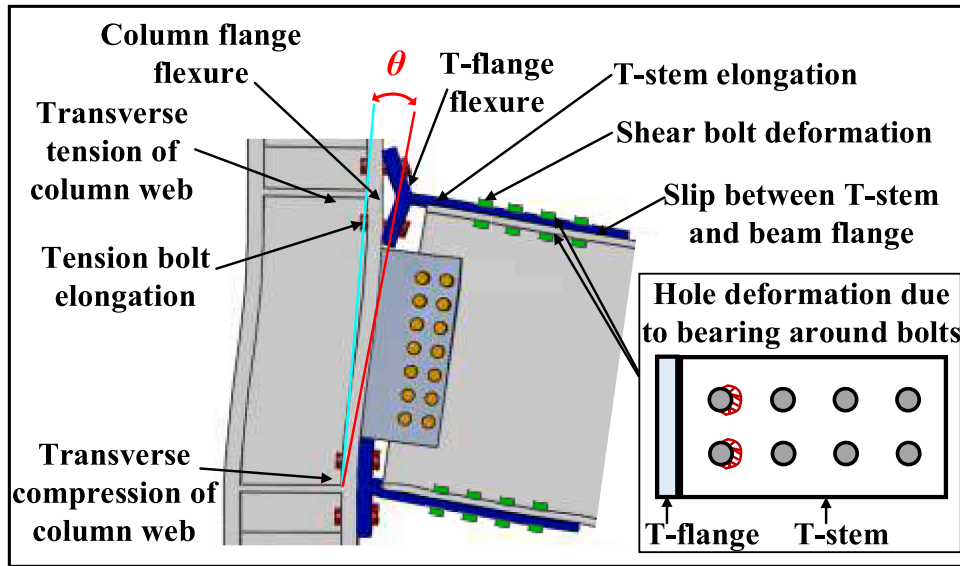


Fig. 3. Illustration of the connection rotation (θ) and component deformations considered for θ .

characterized by moment-rotation M - θ relationships, including the one developed in this study and those provided by ASCE 41-17 [13] and Leon [20]. Here, M_{y-equi} represents equivalent yielding moment; α , β , and c_r are the hardening parameter, post-peak softening parameter, and residual strength parameter, respectively; θ_y represents the rotation corresponding to M_{y-equi} or, if M_{y-equi} is unavailable, the rotation at the endpoint of the initial elastic stage; θ_c and θ_r are the rotation corresponding to M_{max} and $c_r M_{max}$, respectively; θ_u is the ultimate rotation; f_{ye} represents yield strength including material over strength factor; I_b is moment inertia; L_b is beam length; Z_p represents plastic section modulus about major axis.

The cyclic envelope model developed based on the compiled experimental data applies to both PR partial-strength and PR full-strength T-stub connections. It is a multi-linear four-stage cyclic envelope similar to the one provided in ASCE 41-17 [13], but the determination of the most prominent characteristics of the cyclic envelop, e.g., M_{max} , M_{y-equi} , and K_{ini} are different. Specifically, M_{max} is determined by the proposed approach based on different failure modes; M_{y-equi} is deduced from experimental results through the equal-area fitting method; and K_{ini} is determined by the component method that considers all the elastic deformation of the connection components contributing to initial rotational stiffness. Other parameters required to develop the cyclic envelope, including α , β , c_r , and θ_u , are also different to those given by ASCE 41-17 [13].

The cyclic envelope models presented in ASCE 41-17 [13] are characterized by four stages for both PR partial-strength and PR full-strength T-stub connections, but no hardening stage is considered for PR partial-strength T-stub connections. Based on the type of T-stub connections, tabulated values are provided for relevant parameters that control the envelope. Leon [20] also distinguished PR partial-strength and PR full-strength T-stub connections by proposing two different models. Like ASCE 41-17 [13], the characteristic parameters given in [20], e.g., θ_y and θ_c , are determined by assuming empirical constant values for full-strength T-stub connections, or are associated with the connected beam's cross-sectional depth for partial-strength T-stub connections.

4. Determination of the characteristic parameters of the envelope models

The characteristic parameters of the envelope, including M_{max} , M_{y-equi} , K_{ini} , α , β , c_r , and θ_u , determined by the proposed method and other

available methods, are presented in this section. It is worth noting that the parameters M_{y-equi} and α are not applicable to Leon [20]'s partial-strength model.

4.1. Determination of M_{max}

The maximum moment resistance of PR T-stub connections under cyclic loading is a crucial characteristic parameter of the envelope. In addition to the approach proposed in this study, the models provided by ASCE 41-17 [13] and Leon [20] are also presented and evaluated using the experimental data compiled. Additionally, M_{max} determined by EN 1993-1-8:2005 (E) [23] is considered as a reference, even though it is theoretically not applicable for determining the cyclic M_{max} as mentioned earlier.

4.1.1. Failure modes considered

Due to the complexity of T-stub connections, various failure modes can occur. Consequently, M_{max} should be determined by the weakest failure mode controlling the T-stub connection's failure. In the proposed model, a total of five failure modes are considered for PR partial-strength T-stub connections, including T-flange forming plastic mechanism (FM1), T-flange yielding governed by tension bolt fracture due to prying force (FM2), tension bolt fracture without prying force (FM3), T-stem net section fracture (FM4), and shear bolt fracture on the T-stem (FM5). Another failure mode (FM6), which occurs in the case of full-strength T-stub connections, is primarily governed by the formation of a plastic hinge (usually close to the T-stem end) in the connected beam. In the database compiled, the failure modes observed were ductile failure modes of FM1, FM4 and FM6. Brittle fracture failures (FM2, FM3 and FM5), involving high-strength bolt tension and shear fractures, were not reported in the collected literature [9,21,24,35–44].

It is worth mentioning that FM2 is not considered in ASCE41-17 [13], while additional failure modes are considered in Leon's model [20] and EN 1993-1-8:2005 (E) [23], e.g., bearing failure and block tearing failure of the T-stem and local failure of the column flange and beam flange. However, these failure modes are not considered in this study because they rarely govern the connection behavior in practical design. For instance, block tearing failure is usually prevented by limiting the distance from the bolt hole to the plate's edge, while local failure of the column or beam components is typically avoided by limiting the width-to-thickness ratio of the steel plate.

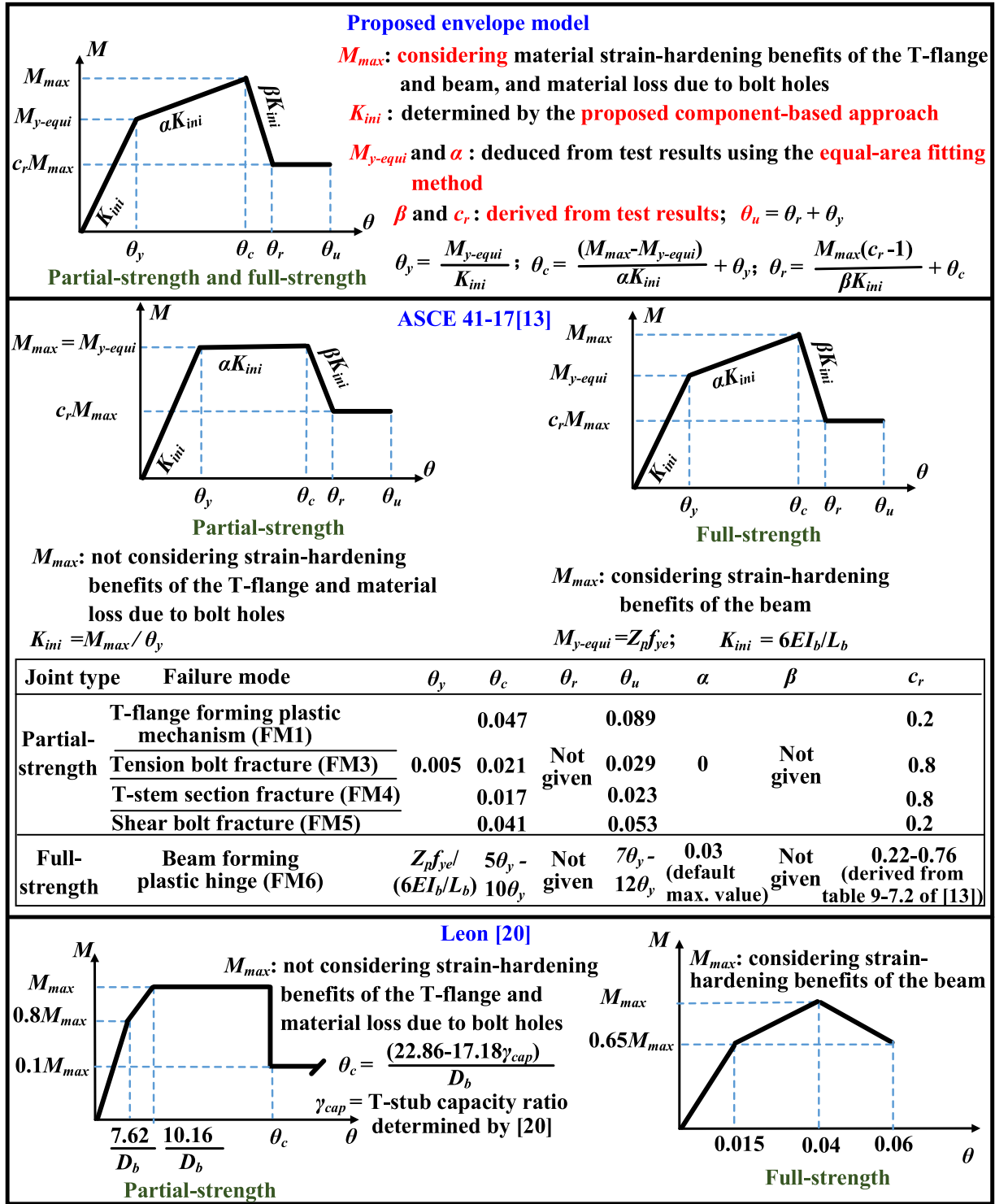


Fig. 4. Different cyclic envelope models for PR full-strength and PR partial-strength T-stub connections.

4.1.2. Equations for calculating M_{max}

The moment resistance (M) in a connection can be represented by a force couple, i.e., a pair of equal but opposite forces (T), applied to the T-stubs. As such, M can be expressed as $M = TZ$, where Z denotes the distance between the center of compression and tension. Consequently, M_{max} of the T-stub connection is determined as $\min\{T_1 Z, \dots, T_6 Z\}$, in which the force T_1 to T_6 correspond to failure mode FM1 to FM6, respectively. It's worth mentioning that different methods employ different lever arms Z , as given in Table 2, with symbols D_b , t_{TS} and m_0

Table 2

The lever arm Z adopted by different methods.

Method	Z
Proposed approach	FM1-FM5: $Z = D_b + t_{TS}$; FM6: $Z = D_b$
EN 1993-1-8:2005 (E) [23]	
Leon [20]	FM1-FM6: $Z = D_b$
ASCE 41-17 [13]	FM1 and FM4: $Z = D_b + t_{TS}$; FM3: $Z = D_b + 2t_{TS} + 2m_0$; FM5 and FM6: $Z = D_b$

denoting the beam cross-sectional depth, T-stem thickness, and the distance from the T-stem side to the centerline of the bolt hole in the first row, respectively.

For the sake of determining the moment resistance for FM1, FM2, FM3, a T-stub is evenly divided into a series of segments, as shown in Fig. 5. Here, n_{tb} is the total number of bolts at the T-flange and l_{Tf} represents the length of the T-flange. Each segment of the T-flange experiences a tension (or compression) force denoted as T' . For FM1, FM2 and FM3, $T_i = (n_{tb}/2)T'_i$ ($i = 1, 2, 3$). As a result, M_{max} is determined by $\min\{(n_{tb}/2)T'_1 z, (n_{tb}/2)T'_2 z, (n_{tb}/2)T'_3 z, T_4 z, T_5 z, T_6 z\}$. The determination of $T'(T)$ for the six failure modes is given below.

- T-flange bending coupled with tension bolt elongation (FM1, FM2 and FM3)

Fig. 6 illustrates the simplified loading models adopted by different methods. A distributed bolt force was employed by Leon [20], which differs from the concentrated bolt force adopted in the proposed approach, ASCE 41-17 [13], and EN 1993-1-8:2005 (E) [23]. Here, Q and B represent prying force and bolt force, respectively. The two parameters, m and n , indicate the position of plastic hinges. They are determined by m_0 , n_0 , d_{tb} , and r , in which n_0 , d_{tb} , and r represent the distance from the T-flange edge to the centerline of the bolt hole, the diameter of the tension bolt shank, and the radius of the K-zone, respectively.

There are three strength models for FM1, FM2, and FM3, corresponding to the simplified loading models mentioned above, as illustrated in Fig. 7. Here, M_p denotes cross-sectional full plastic bending moment, which can be calculated as $M_p = c_{fl} \cdot f_y \cdot (w \cdot t_{Tf}^2)/4$, where c_{fl} is a coefficient accounting for strain-hardening benefits and is taken as 1.16 in the proposed model and neglected in the other three models ([13,20,23]); t_{Tf} denotes T-flange thickness.

Based on the strength models, the expressions for T'_1 , T'_2 and T'_3 for different methods can be determined, as exhibited in Fig. 8. Here, d_{hole} is the bolt hole diameter; f_{ub} represents the ultimate tensile strength of the bolt; A_b is gross area of the bolt shank; and $A_{b,eff}$ is the effective area of the bolt shank ($A_{b,eff} \approx 0.752A_b$ for A490 bolts [22]). The procedure for developing these expressions for the proposed approach is outlined in Appendix B. Similar procedures were employed by ASCE 41-17 [13], EN 1993-1-8:2005 (E) [23] and Leon [20]. Fig. 8 also reports the calculation of T' for FM4, FM5, and FM6, as further detailed below.

- T-stem net section fracture (FM4).

T_4 for FM4 is equal to the ultimate tensile strength of the T-stub

multiplied by the section area considering holes. Both EN 1993-1-8:2005 (E) [23] and ASCE 41-17 [13] employ the same method for calculating the net section area (A_n), which is also adopted by the proposed approach. On the contrary, Leon [20] has adopted the effective net section area ($A_{n,eff} \leq A_n$) as proposed by Swanson [22].

- Shear bolt fracture on the T-stem (FM5).

T_5 for FM5 is equal to the total shear resistance of bolts on the T-stem. The determination of the shear resistance of one bolt varies among different approaches, as shown in Fig. 8. Here, n_{sb} represents the total number of shear bolts at the T-stem; f_{yb} is the nominal shear strength of the bolt; and $A=A_b$ when threads are excluded from the shear plane, otherwise, $A=A_{b,eff}$.

- Beam forming plastic hinge at the T-stem end (FM6)

M_{max} for FM6 is considered as the moment resistance of the beam cross-section. Consequently, T_6 is expressed as the beam plastic bending moment multiplied by a factor to account for strain-hardening benefits and then divided by the beam depth. The proposed approach recommends a factor $c_{be} = 1.16$, while Leon [20] employed a coefficient of 1.1, which leads to slight underestimation. ASCE 41-17 [13] permits a procedure to determine this factor [13], which allows different values. For example, the derived c_{be} ranges from 1.195 to 1.27 for the collected specimens in the compiled database. In contrast, EN 1993-1-8:2005 (E) [23] adopts $c_{be} = 1.0$ without accounting for benefit of strain-hardening.

From the above equations for failure modes, the differences between the proposed approach, Leon [20], ASCE 41-17 [13] and EN 1993-1-8:2005 (E) [23] include the position of plastic hinges (related to m and n), the type of bolt forces (concentrated force or distributed pressure), whether material strain-hardening benefits are considered, whether material loss due to bolt holes is considered, and the calculation of tension resistance (B_u) of bolts.

4.1.3. Evaluation of the predicted M_{max} against experimental results

The accuracy of the M_{max} predicted by the proposed approach, Leon [20], ASCE 41-17 [13] and EN 1993-1-8:2005 (E) [23] is assessed in this section. For the determination of M_{max} , the experimental material strengths were employed. Design partial safety factors in different approaches or design codes, if any, are taken as unity for fair comparison. The specimens that didn't reach their actual maximum capacity are excluded from the assessment of different methods. The comparison of the predicted results against available (29) test results is shown in Fig. 9, in which the predicted M_{max} by the proposed approach, Leon [20], ASCE

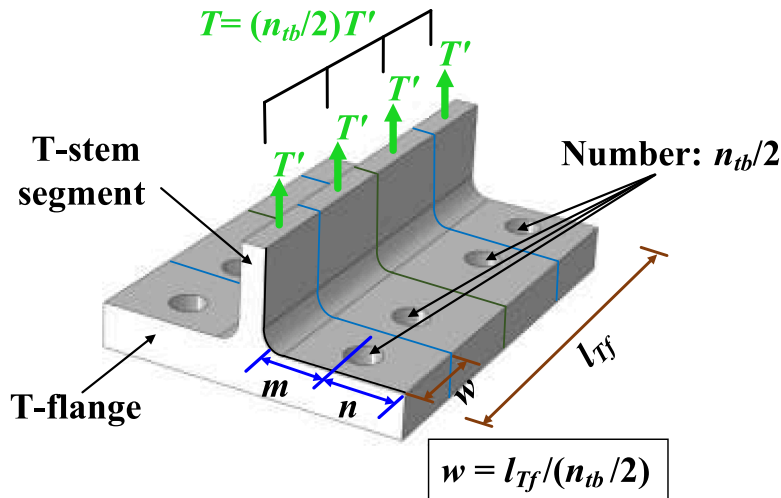


Fig. 5. Division of a T-stub for FM1, FM2 and FM3.

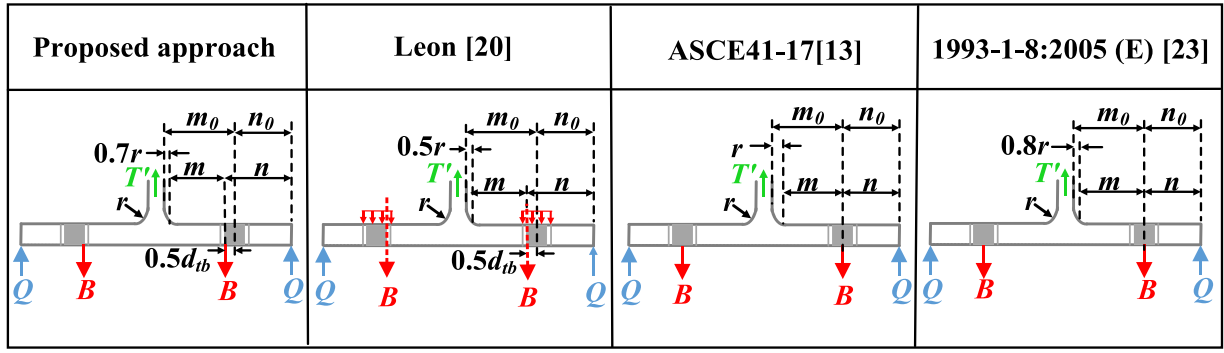


Fig. 6. Simplified loading models adopted by different methods.

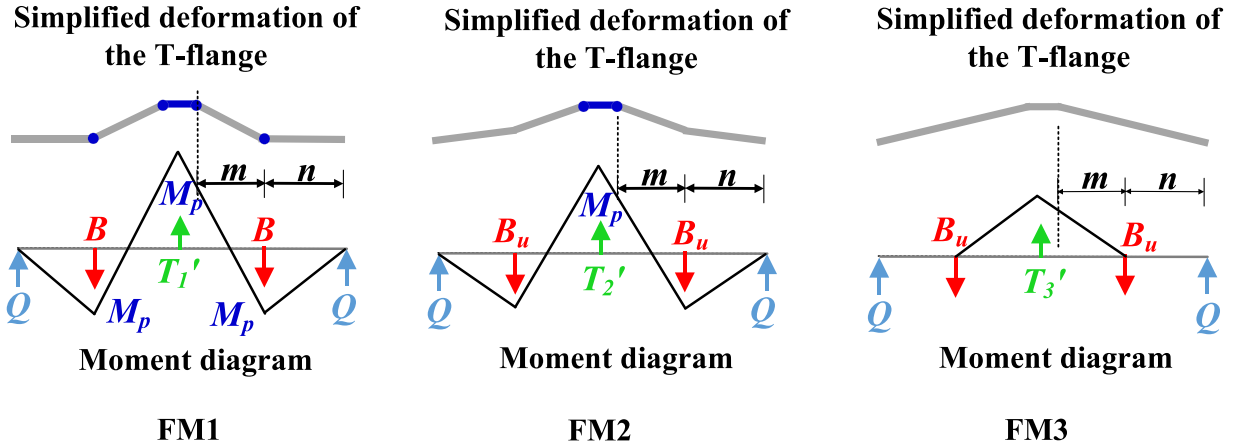


Fig. 7. Strength models for FM1, FM2, and FM3.

Joint type	Failure mode	Proposed approach	Leon [20]	ASCE41-17[13]	1993-1-8:2005 (E) [23]
Partial-strength	FM1	$T'_1 = \frac{(1+\delta)wt_{Tf}^2 c_{fl} f_y}{2m}$ $\delta = 1 - \frac{d_{hole}}{w} \quad c_{fl}=1.16$	$T'_1 = \frac{2(2n - \frac{d_{ib}}{4})wt_{Tf}^2 f_y}{4mn - d_{ib}(m+n)}$		$T'_1 = \frac{wt_{Tf}^2 f_y}{m}$
	FM2	$T'_2 = \frac{\frac{1}{2}wt_{Tf}^2 f_y + 2nB_u}{m+n}$		Not considered	$T'_2 = \frac{\frac{1}{2}wt_{Tf}^2 f_y + 2nB_u}{m+n}$
	FM3	$T'_3 = 2B_u$			
	FM4	$B_u = f_{ub}A_b$	$B_u = (0.75f_{ub})A_b$	$B_u = f_{ub}A_b$	$B_u = (0.9f_{ub})\underline{A_{b,eff}}$
	FM5	$T'_5 = f_{vb}n_{sb}A_b$			
Full-strength	FM6	$T_6 = \frac{c_{be}Z_p f_y}{D_b}$ $c_{be}=1.16$	$T_6 = \frac{1.1Z_p f_{ye}}{D_b}$	$T_6 = \frac{c_{be}Z_p f_{ye}}{D_b}$ $1.2 \leq c_{be} \leq 1.27$	$T_6 = \frac{Z_p f_y}{D_b}$
		$f_{vb} \text{ is given in Table 5.1 of RCSC [47]}$	$f_{vb} \text{ is given in AISC-LRFD [50]}$	$f_{vb} \text{ is given in AISC-LRFD [50]}$	$f_{vb} \text{ is given in Table 3.4 of [23]}$

Fig. 8. Comparison of equations for calculating T (T') provided by different methods.

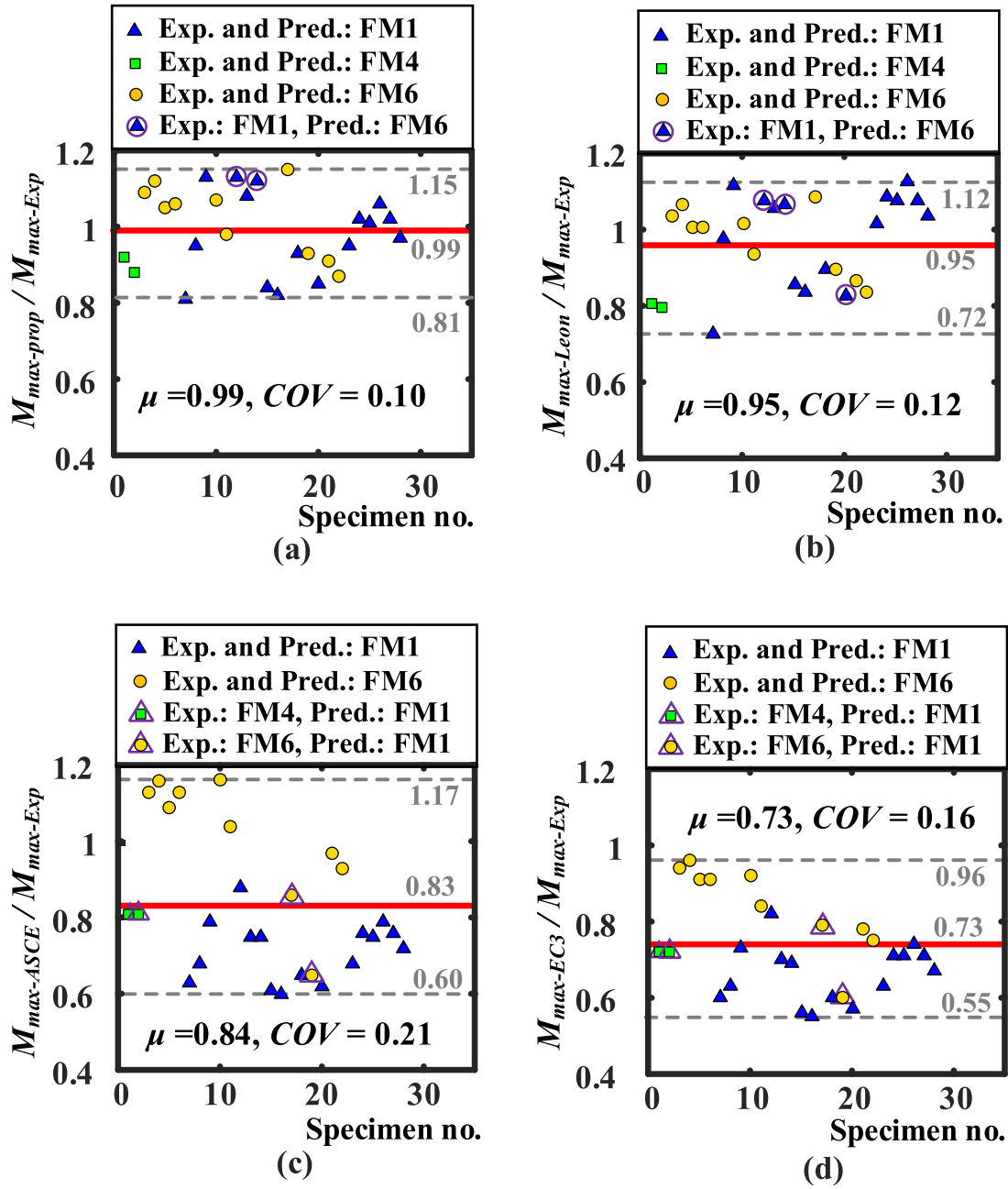


Fig. 9. Comparison of predicted M_{max} by (a) proposed approach, (b) Leon [20], (c) ASCE 41-17 [13], and (d) EN 1993-1-8:2005 (E) [23] against experimental results.

41-17 [13] and EN 1993-1-8:2005 (E) [23] are denoted by $M_{max-prop}$, $M_{max-Leon}$, $M_{max-ASCE}$, and $M_{max-EC3}$, respectively, and the experimental M_{max} is denoted by $M_{max-Exp}$. The vertical axis represents the ratio of predicted to experimental M_{max} , and the horizontal axis represents the specimen number (1, 2, 3, ...). Additionally, the average ratio (μ) is shown as the horizontal solid line, together with the maximum and minimum prediction errors, as indicated by the dashed lines.

In Fig. 9, failure modes exhibited when reaching M_{max} in the tests are distinguished by different markers. It is observed that the predicted failure modes by the proposed approach for all the available specimens match the experimental results, except the two specimens reported in [40]. For the two specimens failed in FM1, the proposed approach predicts FM6. This discrepancy arises because the ultimate strength of the T-flange is very close to that of the beam for the two specimens. By contrast, Leon [20]'s model for three specimens is inconsistent with the test results, while both ASCE 41-17 [13] and EN 1993-1-8:2005 (E) [23]

have four predicted failure modes that do not match the experimental results.

It is observed that the proposed approach yields the most reasonable results, exhibiting a μ of 0.99 and a coefficient of variation (COV) of 0.10. Its estimation errors range from -19% to $+15\%$. By contrast, Leon [20]'s method gives a maximum underestimation error of -28% and a μ of 0.95, while ASCE 41-17 [13] generates a maximum underestimation error of -40% . It is worth mentioning that EN 1993-1-8:2005 (E) [23] is overly conservative in predicting the resistance of T-stub connections because the predicted M_{max} for the monotonic backbone is still much lower than the cyclic experimental values for the test specimens considered.

The improved accuracy of the proposed model can be attributed to several factors. Specifically, for FM1, the improvement is mainly due to (1) the explicit consideration of material strain-hardening benefits and the material loss due to bolt holes, which are both neglected in Leon

[20], EN 1993-1-8:2005 (E) [23] and ASCE 41-17 [13], and (2) the adopted $m = m_0 - 0.7r - d_{th}/2$, compared to $m = m_0 - 0.8r$ and $m_0 - r$, respectively, in [23] and [13]. For FM6, which occurred in full-strength T-stub connections, the improvement is mainly due to the appropriate value ($c_{be} = 1.16$) to consider the beneficial effect of strain hardening. The large values c_{be} (ranging from 1.2 to 1.27 for the specimens considered) according to ASCE 41-17 [13] overestimates the strain-hardening benefits, leading to overestimated M_{max} for FM6. In contrast, the EN 1993-1-8:2005 (E) [23] underestimates M_{max} mainly due to the neglect of strain hardening effects. Regarding FM4, despite only two specimens exhibiting this failure mode, the predictions of the proposed approach still outperform those of other methods.

4.2. Determination of M_{y-equi}

M_{y-equi} is determined using the equal-area fitting method [49], as depicted in Fig. 10. A summary of the deduced M_{y-equi} values is shown in Table 3. It should be pointed out that M_{y-equi} is not applicable to the monotonic model given in EN 1993-1-8:2005 (E) [23] and the partial-strength cyclic model provided by Leon [20]. Specimens that didn't reach their actual maximum capacity were excluded from the M_{y-equi} deduction, as M_{max} is unavailable for them. Based on those deduced from the experimental data, M_{y-equi} can be best estimated as the average value between $0.7M_{max-Exp}$ and $0.73M_{max-Exp}$. As such, $M_{y-equi} = 0.72M_{max-prop}$ is adopted in the proposed model.

4.3. Determination of K_{ini}

The component method is adopted in the proposed approach to determine K_{ini} for PR T-stub connections. This section provides insights into the components considered within this approach and outlines the procedure for calculating K_{ini} using the proposed method. Additionally, it highlights the distinctions between the proposed approach and other existing methods, including ASCE 41-17 [13], Leon [20], EN 1993-1-8:2005 (E) [23], Swanson [22], and Wang [25]. The K_{ini} values obtained through various methods are also assessed using the experimental data compiled.

4.3.1. Components considered by the proposed approach for determining K_{ini}

The initial rotational stiffness (K_{ini}) of a connection can be determined assuming its components remain in the elastic stage. The proposed approach considers all the components contributing to the deformation of a T-stub connection, as illustrated in Fig. 11. These components include those contributed by column web in transverse compression ($\Delta_{cw,c}$) in case of no web stiffener, column web in transverse tension ($\Delta_{cw,t}$) in case of no web stiffener, column flange in bending (Δ_{cf}) due to tension force, T-flange in bending (Δ_{Tf}) due to

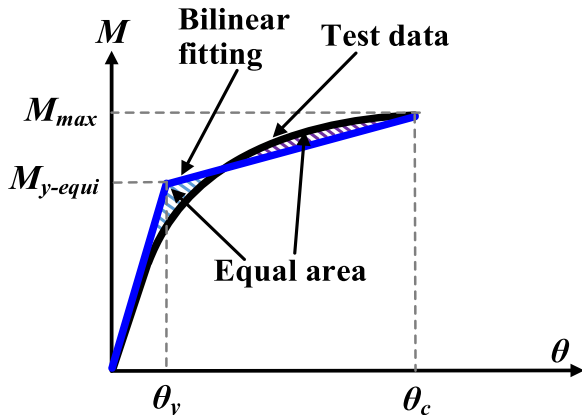


Fig. 10. Illustration of the equal-area fitting method.

Table 3

Summary of M_{y-equi} determined by different methods.

Method	PR T-stub connection type	M_{y-equi}
Deduced by equal-area fitting	Partial-strength	$\mu = 0.7M_{max-Exp}$ and $COV = 0.16$; Ranges from 0.6 to $0.84 M_{max-Exp}$
	Full-strength	$\mu = 0.73M_{max-Exp}$ and $COV = 0.11$; Ranges from 0.58 to 0.87 $M_{max-Exp}$ $0.72 M_{max-prop}$
Proposed simplified approach Leon [20]	Partial-strength	Not applicable
	Full-strength	$0.65M_{max-Leon}$
ASCE41-17 [13]	Partial-strength	$1.0M_{max-ASCE}$
	Full-strength	Z_{pfe} (around $0.79 \sim 0.83M_{max-ASCE}$)

tension force, T-stem in tension (Δ_{Ts}), bolts in tension ($\Delta_{bol,t}$), bolts in shear ($\Delta_{bol,s}$), T-stem hole deformation due to bearing around bolts ($\Delta_{Ts,be}$), beam flange hole deformation due to bearing around bolts ($\Delta_{bf,be}$), and slip between the T-stem and beam flange (Δ_{slip}).

The components considered in this study are comprehensive compared to those considered by EN 1993-1-8: 2005(E) [23] and Swanson [22], as illustrated in Fig. 12. The limitations within [23] restrict its coverage to certain connection types, thereby excluding the consideration of T-stem elongation (Δ_{Ts}) in tension and slip (Δ_{slip}) between the T-stem and beam flange. In contrast, Swanson [22]'s stiffness model ignores column flange deformation and column web deformation.

4.3.2. Development of the expression for K_{ini} in the proposed approach

The expression for K_{ini} is developed based on the components considered, and the detailed procedure is outlined in Fig. 13. Note that the relationship between the force T and elastic deformation for each component (denoted as Δ_i) is established as $T = (Ek_i)\Delta_i$, where E represents Young's modulus and k_i (with units in mm) is stiffness coefficient of the component. By replacing each Δ_i with $T/(Ek_i)$, the expression of K_{ini} is obtained, given by

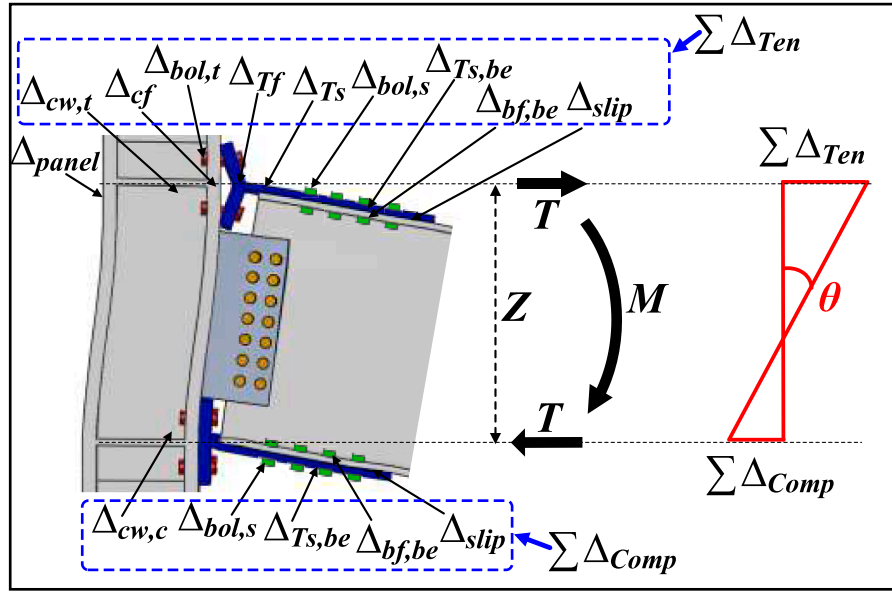
$$K_{ini} = EZ^2 / \left(\sum (1/k_{Ten}) + \sum (1/k_{Comp}) \right) \quad (1)$$

or

$$K_{ini} = EZ^2 / \sum (1/k_i) \quad (2)$$

where $\sum 1/k_{Ten}$ is the sum of $1/k_i$ for the components in tension; $\sum 1/k_{Ten} = 1/k_{cw,t} + 1/k_{cf} + 1/k_{Tf} + 1/k_{Ts} + 1/k_{bol,t} + 1/k_{bol,s} + 1/k_{Ts,be} + 1/k_{bf,be} + 1/k_{slip}$; $k_{cw,t}$, k_{cf} , ..., k_{slip} corresponds to $\Delta_{cw,t}$, Δ_{cf} , ..., Δ_{slip} , respectively; $\sum 1/k_{Comp}$ is the sum of $1/k_i$ for the components in compression; $\sum 1/k_{Comp} = 1/k_{cw,c} + 1/k_{bol,s} + 1/k_{Ts,be} + 1/k_{bf,be} + 1/k_{slip}$; $k_{cw,c}$, $k_{bol,s}$, ..., k_{slip} corresponds to $\Delta_{cw,c}$, $\Delta_{bol,s}$, ..., Δ_{slip} , respectively.

In the proposed approach, the determination of k_i for relevant components, except for Δ_{Ts} and Δ_{slip} , aligns with the provisions outlined in section 6.3 of EN 1993-1-8:2005 (E) [23]. Detailed formulations for these coefficients are readily available in [23]. Due to the presence of pretension force, it is assumed that no slip occurs between the T-stem and beam flange in the initial loading stage. Consequently, the stiffness coefficient k_i for Δ_{slip} is considered to be infinite. This assumption finds support in evidence derived from the cyclic tests on six full-scale T-stub connections, as reported by Smallidge [21]. For the calculation of k_i for Δ_{Ts} , denoted as k_{Ts} , it is derived according to Swanson [22]. The derived k_{Ts} is given by

Fig. 11. Components considered for the determination of K_{ini} .

Method	Components considered for the determination of K_{ini}
Proposed approach	$\Delta_{cw,c}, \Delta_{cw,t}, \Delta_{cf}, \Delta_{Tf}, \Delta_{bol,t}, \Delta_{bol,s}, \Delta_{Ts,be}, \Delta_{bf,be}, \Delta_{slip}, \Delta_{Ts}$
EN 1993-1-8:2005 (E) [23]	$\Delta_{cw,c}, \Delta_{cw,t}, \Delta_{cf}, \Delta_{Tf}, \Delta_{bol,t}, \Delta_{bol,s}, \Delta_{Ts,be}, \Delta_{bf,be}$
Swanson[20]	$\Delta_{Tf}, \Delta_{bol,t}, \Delta_{Ts,be}, \Delta_{bf,be}, \Delta_{slip}, \Delta_{Ts}$

Fig. 12. Comparison of the components considered for determining K_{ini} by different methods.

$$k_{Ts} = \frac{4l_s t_{Ts} (\tan \theta_{eff})^2}{2l_s \tan \theta_{eff} + g_s \ln \left(\frac{g_s}{l_{eff}} \right)} \quad (3)$$

in which

$$l_{eff} = \min \{ (2l_s \tan \theta_{eff} + g_s), l_{Tf} \} \quad (4)$$

where l_s is the length between the last bolt row and the first bolt row, θ_{eff} denotes the effective angle, and g_s is the distance between two rows of bolt holes. The definitions of l_s , θ_{eff} , and g_s are illustrated in Fig. 14.

4.3.3. Comparison of different methods in determining K_{ini}

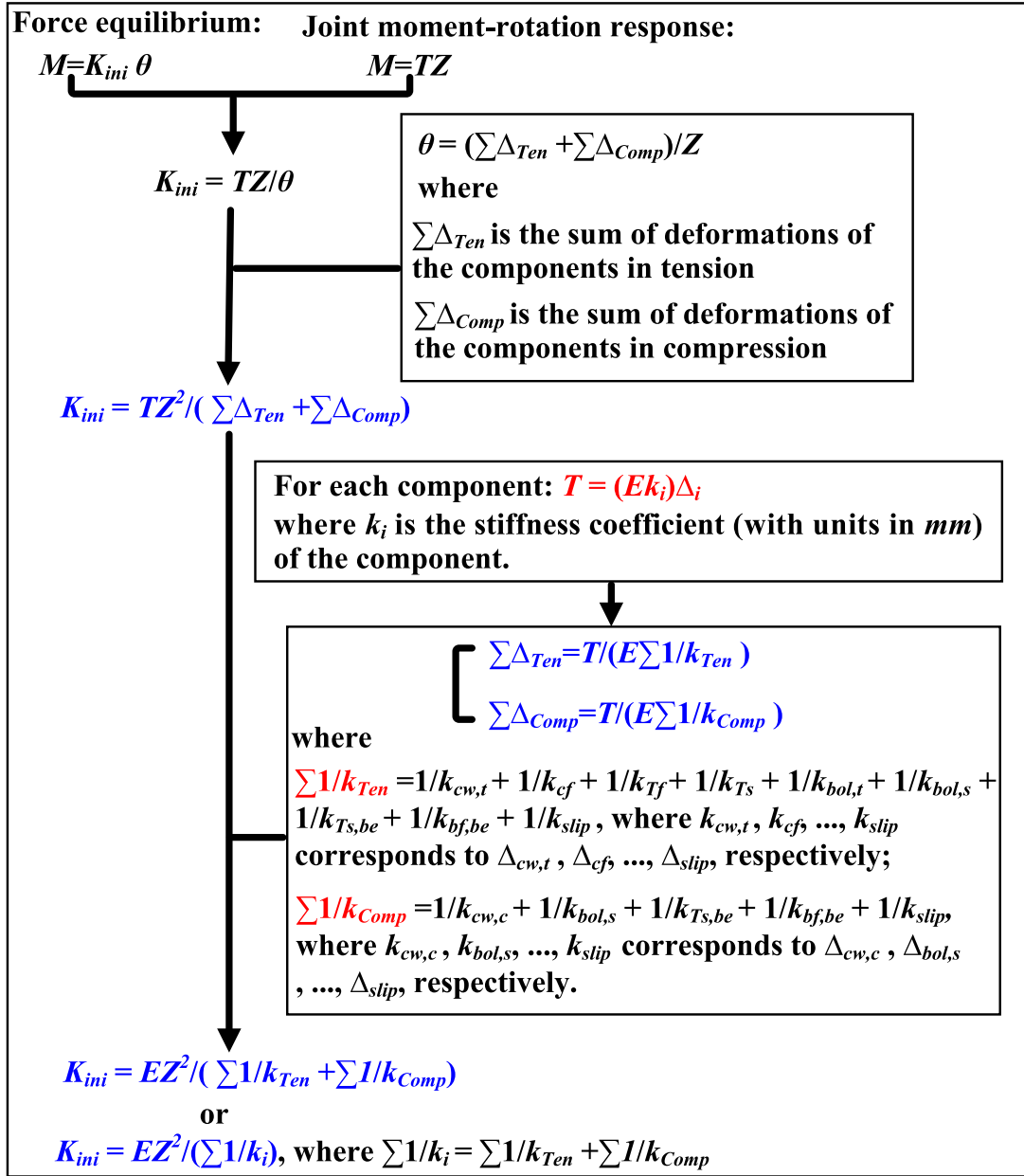
Fig. 15 summarizes the comparison between the proposed approach and EN 1993-1-8:2005 (E) [23] and Swanson [22] when employing the component method to determine the initial rotational stiffness (K_{ini}) of a T-stub connection. Additionally, K_{ini} determined by other available approaches without using the component method, including ASCE 41-17 [13], Leon [20] and Wang [25], are also given in this figure. Note that in ASCE 41-17 [13], K_{ini} for partial-strength and full-strength T-stub connections is determined empirically by $M_{max}/0.005$ and $6EI_b/L_b$, respectively. Similarly, the formula for determining K_{ini} given by Leon [20] for partial-strength and full-strength T-stub connections is

$0.8M_{max}/(7.62/D_b)$ and $0.65M_{max}/0.015$, respectively. Wang [25]'s simplified equation is less empirical, primarily based on the geometry and elastic flexural stiffness of the T-flange.

4.3.4. Evaluation of the predicted K_{ini} against experimental results

The accuracy of the predicted K_{ini} determined by the proposed approach, EN 1993-1-8:2005 (E) [23], Swanson [22], ASCE 41-17 [13], Leon [20] and Wang [25], is evaluated in this section, and the corresponding K_{ini} is denoted by $K_{ini-prop}$, $K_{ini-EC3}$, $K_{ini-Swanson}$, $K_{ini-ASCE}$, $K_{ini-Leon}$ and $K_{ini-Wang}$, respectively. Design partial safety factors in different approaches, if any, are taken as unity. Note that the total number of specimens used for evaluating the predicted K_{ini} is 35, which differs from that (29) used for evaluating the predicted M_{max} . Comparison of the predicted results by different approaches against available experimental results is shown in Fig. 16, in which the experimental K_{ini} is termed $K_{ini-Exp}$, and the vertical axis represents the ratio of the predicted K_{ini} to $K_{ini-Exp}$ while the horizontal axis represents the specimen number. For EN 1993-1-8:2005 (E) [23] and ASCE 41-17 [13], the predicted results for full-strength and partial-strength T-stub connections are distinguished by different signs, as different formulas were employed for full-strength and partial-strength connections.

It is observed that the proposed approach demonstrates the superior

Fig. 13. Detailed procedure for the development of K_{ini} .

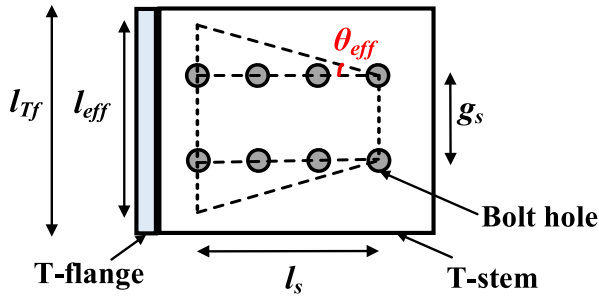
accuracy, with a μ of 1.02 and a COV of 0.15. The estimation errors vary from -26% to $+23\%$, indicating a precise prediction of K_{ini} . While the distribution of the predicted results by EN 1993-1-8:2005 (E) [23] is similar to $K_{ini-prop}$, $K_{ini-EC3}$ is less accurate, as evidenced by a μ of 1.12, a COV of 0.17. This is because the T-stem in tension (Δ_{Ts}) is not considered by $K_{ini-EC3}$. Note that Swanson [22] tends to overestimate the K_{ini} for most specimens, in which the maximum overestimation error reaches $+90\%$. This is attributed to neglecting the influence of column flange deformation and column web deformation on the initial rotational stiffness.

The predictions from ASCE 41-17 [13] exhibit large variability, with estimation errors ranging from -48% to $+165\%$. This indicates that the empirical rotation value of 0.005 used for partial-strength connections and the 6 times bending stiffness ($6EI_b/L_b$) of the beam for full-strength connections should be improved to account for diverse material and geometric properties. Leon [20]'s approach significantly underestimates the actual K_{ini} , with a μ of 0.38 for $K_{ini-Leon}/K_{ini-Exp}$. All the predicted K_{ini} values are no more than 50 % of the experimental results. The notable

underestimation is primarily caused by the relatively large yielding rotation value, which is 0.015 for full-strength connections and $7.62/D_b$ (ranges from 0.0145 to 0.0305) for partial-strength connections considered. The simplified equation proposed by Wang [25] can overestimate $K_{ini-Exp}$ up to 450 %. This overestimation is attributed to the limited applicability of Wang [25]'s approach, which is suitable only for specific material strength and geometric configurations.

4.4. Determination of other parameters for the cyclic envelope

In addition to K_{ini} , M_{y-equi} and M_{max} , the cyclic envelope's construction requires other parameters: α , β , c_r and θ_u , in which α can be determined simultaneously with M_{y-equi} using the equal-area fitting method. As for the parameter β , it is derived by $(M_{min-Exp} - M_{max-Exp})/(\theta_{min-Exp} - \theta_{c-Exp})/K_{ini-Exp}$. Here, $M_{min-Exp}$ is experimental minimum moment in the post-peak descending branch and its corresponding rotation is $\theta_{min-Exp}$. The mean values of α and β for PR partial-strength T-stub connections are 0.047 and -0.18 , respectively, and 0.043 and



if $t_{Ts} \leq 12.7\text{mm}$, $\theta_{eff} = 30^\circ$;
 if $6.35\text{mm} \leq t_{Ts} < 12.7\text{mm}$, $\theta_{eff} = 2.36(^\circ/\text{mm}) * t_{Ts}$;
 if $t_{Ts} < 6.35\text{mm}$, $\theta_{eff} = 15^\circ$.

Fig. 14. Definition of relevant geometric parameters for determining k_{Ts} [22].

–0.16 for PR full-strength T-stub connections, respectively. Note that limited data are available for c_r and θ_u . Thus, recommendations are made based on the experimental observations as follows. The residual strength parameter, c_r , is derived from the specimen with the lowest post-peak unloading point. In case of partial-strength T-stub

connections, c_r equals 0.59. For full-strength T-stub connections, c_r is 0.63. Since none of the collected specimens reached the eventual residual strength stage, θ_u is not able to be derived from experimental results directly. In this study, θ_u is approximately taken as $\theta_r + 2\theta_y$ for both partial-strength and full-strength T-stub connections. Table 4 provides an overview of the adopted α , β , c_r and θ_u for the proposed approach, along with those provided by Leon [20] and ASCE 41-17 [13].

5. Comparison of different envelope models against experimental results

The developed cyclic envelope model is evaluated by comparing it to the experimental moment-rotation envelopes, with data points taken at each peak displacement during the first cycle of each loading increment. Additionally, the cyclic envelope models provided by Leon [20] and ASCE 41-17 [13] are also assessed using experimental data. It's worth noting that ASCE 41-17 [13] doesn't specify the slope of the post-peak branch of the envelope model. To make the predicted results by [13] comparable to those obtained from the proposed envelope, the parameter β in [13] is considered to be the same as that in the proposed model.

The comparison of different envelope models against experimental results for the available (23) specimens is presented in Fig. 17. The assessment shows that Leon's model [20] yields unsatisfactory

Method	Determination of K_{ini}		
Proposed approach		each k_i is determined by Table 6.11 provided in [23] determined by EQ 7-32 provided in [22] $\sum 1/k_{Ten} = 1/k_{cw,t} + 1/k_{cf} + 1/k_{Tf} + 1/k_{bol,t} + 1/k_{bol,s} + 1/k_{Ts,be} + 1/k_{bf,be} + 1/k_{slip} + 1/k_{Ts}$ $\sum 1/k_{Comp} = 1/k_{cw,c} + 1/k_{bol,s} + 1/k_{Ts,be} + 1/k_{bf,be} + 1/k_{slip}$ taken as infinity for preloaded bolts	$Z = D_b + t_{Ts}$
EN 1993-1-8:2005 (E) [23]	$K_{ini} = EZ^2 / (\sum 1/k_{Ten} + \sum 1/k_{Comp})$	each k_i is determined by Table 6.11 provided in [23] $\sum 1/k_{Ten} = 1/k_{cw,t} + 1/k_{cf} + 1/k_{Tf} + 1/k_{bol,t} + 1/k_{bol,s} + 1/k_{Ts,be} + 1/k_{bf,be}$ $\sum 1/k_{Comp} = 1/k_{cw,c} + 1/k_{bol,s} + 1/k_{Ts,be} + 1/k_{bf,be}$	$Z = D_b + t_{Ts}$
Swanson [22]		determined by EQ 7-32 provided in [22] $\sum 1/k_{Ten} = \frac{1/k_{Tf} + 1/k_{bol,t} + 1/k_{Ts,be} + 1/k_{bf,be} + 1/k_{slip} + 1/k_{Ts}}{1/k_{Ts,be} + 1/k_{bf,be} + 1/k_{slip}}$ treated together, determined by EQ 9-40 provided in [22] treated together, determined by EQ 9-38 provided in [22]	$Z = D_b$
ASCE 41-17 [13]	Partial-strength: $K_{ini} = M_{max} / 0.005$; Full-strength: $K_{ini} = 6EI_b / L_b$		
Leon [20]	Partial-strength: $K_{ini} = 0.8M_{max} / (7.62/D_b)$; Full-strength: $K_{ini} = 0.65M_{max} / 0.015$		
Wang [25]	$K_{ini} = 192EI_T \cdot Z^2 / (e^3 + 12.48t_{Tf}^2 \cdot e)$, where $EI_T = 1/12 \cdot w \cdot t_{Tf}^3$; $e = 2m_0 + t_{Ts}$; $Z = D_b - t_{bf}$		

Fig. 15. Comparison between various methods in determining K_{ini} for T-stub connections.

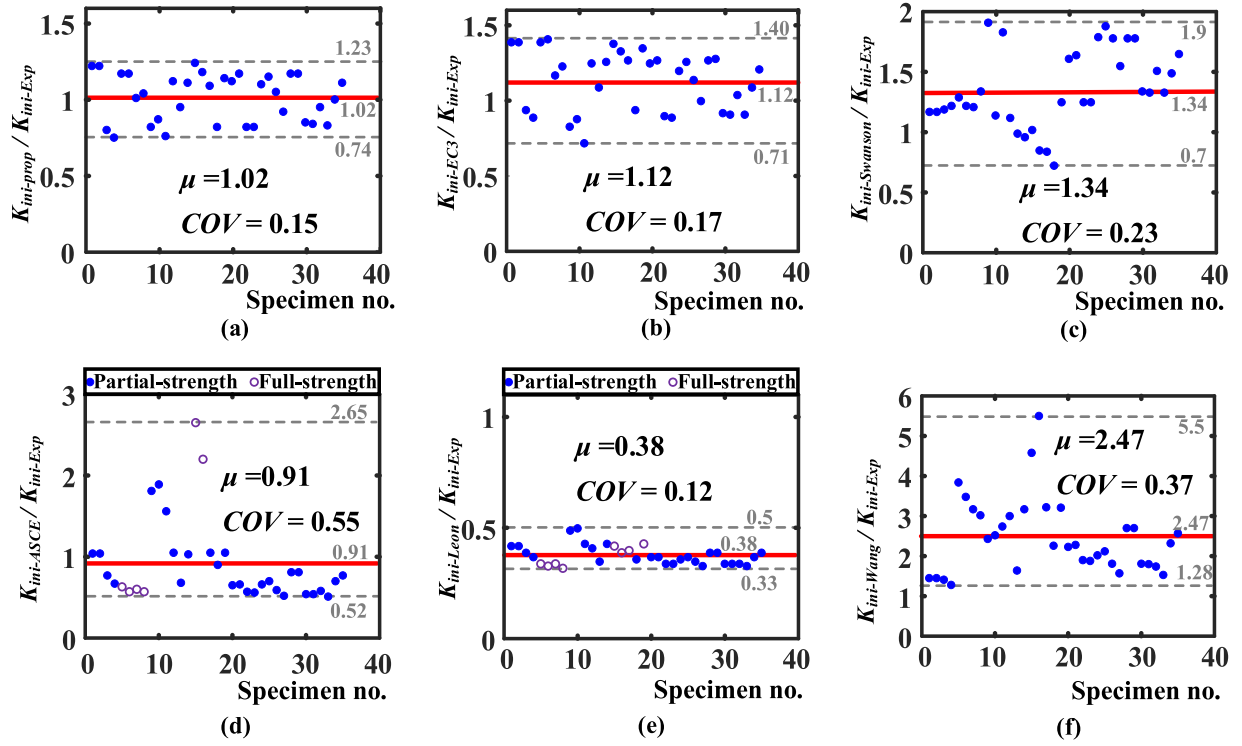


Fig. 16. Comparison of the predicted K_{ini} by (a) the proposed approach, (b) EN 1993-1-8:2005 (E) [23], (c) Swanson [22], (d) ASCE 41-17 [13], (e) Leon [20], and (f) Wang [25] against experimental results.

Table 4
Summary of α , β , c_r and θ_u provided by different methods.

Method	T-stub connection type	Failure mode	α	β	c_r	θ_u
Proposed approach	Partial-strength	FM1-FM5	0.047 (Mean value deduced from test results; $COV=0.33$)	-0.18 (Mean value deduced from test results; $COV=-0.21$)	0.59 (Derived from specimen TS-CYC 04)	$\theta_r + 2\theta_y$ (0.0927 for specimen TS-CYC 04)
	Full-strength	FM6	0.043 (Mean value deduced from test results; $COV=0.24$)	-0.16 (Mean value deduced from test results; $COV=-0.15$)	0.63 (Derived from specimen T2)	$\theta_r + 2\theta_y$ (0.086 for specimen T2)
Leon [20]	Partial-strength	FM1-FM5	Not applicable	Negative infinity	0.10	Not given
	Full-strength	FM6	$0.35M_{max-Leon}/0.02$ ($\mu=0.052$, $COV=0.28$)	$-0.35M_{max-Leon}/0.02$ ($\mu=-0.086$, $COV=-0.34$)	0.65	0.06
ASCE41-17 [13]	Partial-strength	FM1	0	Not given	0.2	0.089
		FM3			0.8	0.029
		FM4			0.8	0.023
		FM5			0.2	0.053
	Full-strength	FM6	0.03 (Default maximum value)		$0.436-0.76^{(1)}$	$9.5\theta_y-12\theta_y$ ($\mu=0.056$, $COV=0.31$)

Note: ⁽¹⁾: these values are applicable to the tested specimens compiled.

predictions for the T-stub connections compiled, primarily due to an excessively low initial rotation stiffness resulting from an overestimation of yielding rotation. Moreover, a noticeable discrepancy exists between the post-peak branch of Leon's model [20] and the experimental envelope, particularly for PR partial-strength T-stub connections. Clearly, the performance of ASCE 41-17 [13] is unstable, as it tends to significantly underestimate the envelope for PR partial-strength specimens, such as PJD1-N and JATD12-N. In contrast to Leon's [20] model and ASCE 41-17 [13], the proposed envelope model consistently produces reasonable predictions for the collected specimens. From Fig. 17, it can be concluded that the proposed envelope model demonstrates good accuracy and robustness for the three failure modes: T-flange forming plastic mechanism (FM1), T-stem net section fracture (FM4), and the formation of a plastic hinge in the connected beam (FM6).

Additionally, the metric "relative root mean square error (RRMSE)" is used to quantify the prediction error. RRMSE is defined as

$$RRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (M_{i,Exp} - M_{i,pred})^2}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (M_{i,Exp})^2}} \quad (5)$$

where $M_{i,pred}$ and $M_{i,Exp}$ represent moment determined by the proposed and experimental envelope, respectively. n is the total number of data point considered in the envelope. To calculate RRMSE, the cyclic envelope is discretized to generate a large number of points, as depicted in Fig. 18. A lower RRMSE indicates a better fit of the envelope model to the experimental data. Conversely, a higher RRMSE indicates a less accurate model with larger errors.

The RRMSE values for the different models are presented in Table 5. Compared to both ASCE 41-17 [13] and Leon's model [20], the proposed model not only shows lower average RRMSE but also displays less variation in RRMSE across different specimens. These results reinforce the proposed model's good accuracy and robustness, affirming its

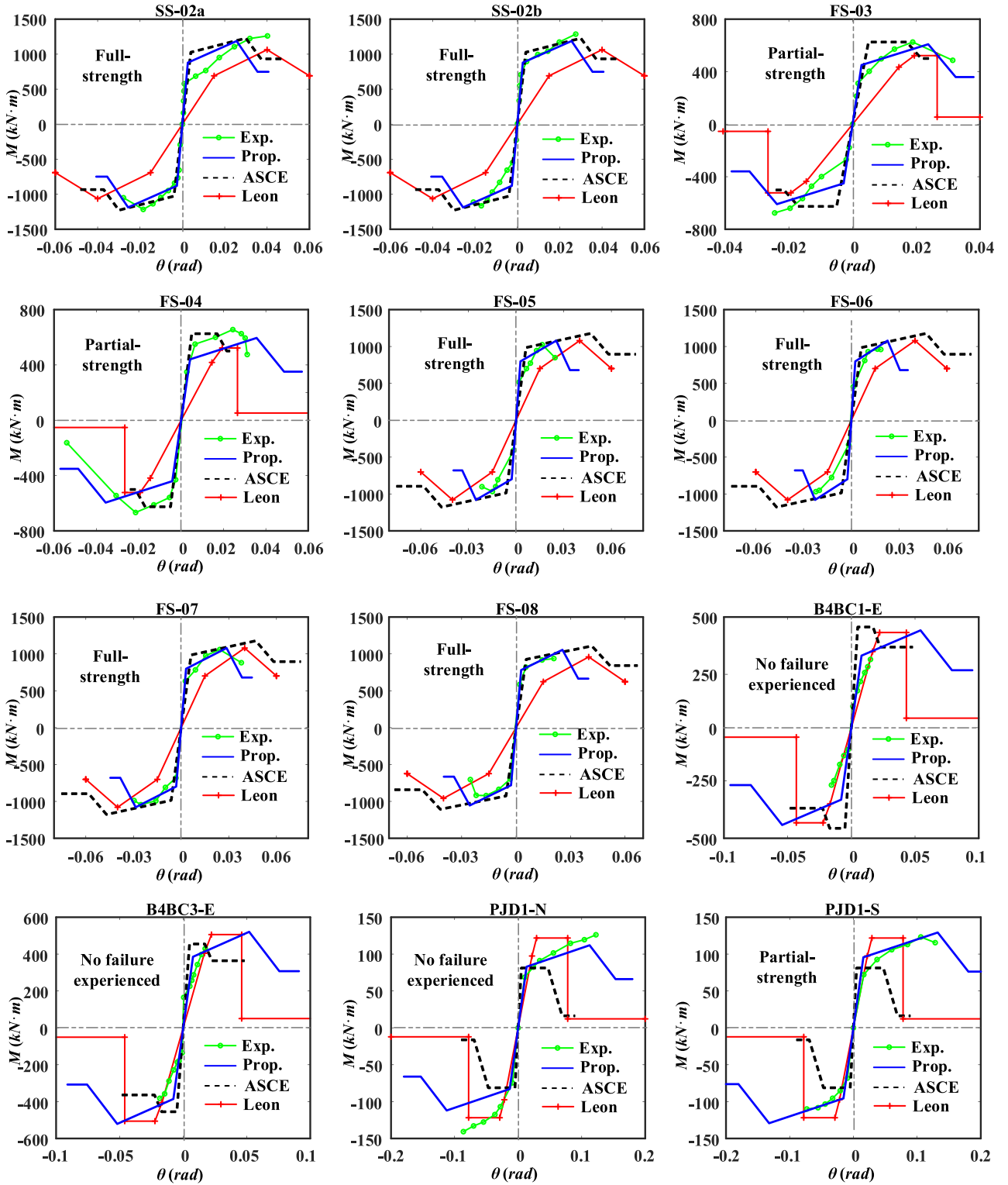


Fig. 17. Comparison of different envelope models against experimental results.

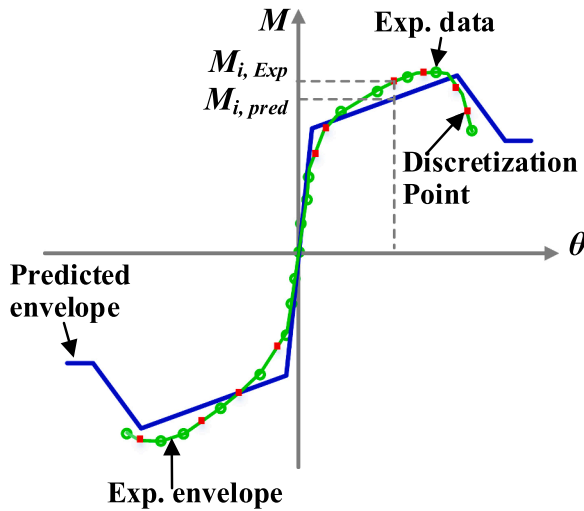


Fig. 18. Discretization of the envelope for the calculation of RRMSE.

Table 5
RRMSE for different models.

Specimen	Classification	RRMSE		
		Proposed model	ASCE41-17 [13]	Leon [20]
SS-02a	Full-strength	0.189	0.167	0.384
SS-02b	Full-strength	0.100	0.108	0.379
FS-03	Partial-strength	0.149	0.504	0.393
FS-04	Partial-strength	0.188	0.599	0.555
FS-05	Full-strength	0.119	0.274	0.391
FS-06	Full-strength	0.093	0.193	0.256
FS-07	Full-strength	0.077	0.228	0.272
FS-08	Full-strength	0.161	0.173	0.401
B4BC1-E	No failure experienced	0.234	0.840	0.107
B4BC3-E	No failure experienced	0.131	0.327	0.119
PJD1-N	No failure experienced	0.170	0.718	0.519
PJD1-S	Partial-strength	0.065	0.693	0.860
PJD2-N	No failure experienced	0.168	0.743	0.671
PJD2-S	No failure experienced	0.182	0.754	0.662
KJD1-N	Partial-strength	0.118	0.621	0.626
KJD1-S	Partial-strength	0.146	0.600	0.589
KJD2-N	No failure experienced	0.145	0.829	0.756
JATD12-N	Partial-strength	0.379	0.887	0.660
JATD12-S	Partial-strength	0.153	0.616	0.572
JATD14-N	Partial-strength	0.110	0.445	0.201
JATD14-S	No failure experienced	0.161	0.434	0.345
Specimen1	No failure experienced	0.088	0.254	0.560
Specimen 2	Partial-strength	0.163	0.330	0.533
	Mean	0.148	0.504	0.458
	Standard deviation	0.061	0.239	0.192

effectiveness in capturing the intricate behavior of PR T-stub connections under cyclic loading conditions.

6. Conclusions

The current study focuses on developing a reliable moment-rotation envelope for PR T-stub connections subjected to cyclic loading. The main conclusions drawn are as follows:

- (1) The proposed approach for determining M_{max} demonstrates superior accuracy compared to various existing methods, including those from Leon [20], ASCE 41-17 [13], and EN 1993-1-8:2005 (E) [23]. The enhanced precision for FM1 is primarily due to the explicit consideration of material strain-hardening benefits and the material loss due to bolt holes, and the adoption of an appropriate geometric parameter (m). The improved accuracy for FM6 is attributed to the use of an appropriate coefficient that accounts for beam strain-hardening benefits.
- (2) In terms of determining K_{ini} , the proposed approach effectively addresses the limitations present in other methods, such as EN 1993-1-8:2005 (E) [23] and Swanson [22]. The proposed approach yields the most accurate K_{ini} , outperforming EN 1993-1-8:2005 (E) [23], while Swanson [22]'s method tends to overestimate experimental K_{ini} . Empirical equation-based methods, such as ASCE 41-17 [13] and Leon [20], exhibit noticeable variability and inaccuracy, respectively. Wang [25]'s simplified equation demonstrates significant overestimation, primarily due to its limited applicability.
- (3) M_{y-equi} is estimated to be 0.72 times the predicted maximum moment, based on experimental data. Additionally, key envelope parameters, including α , β , and c_r , are deduced from test data.
- (4) Comparison of different envelope models against experimental data demonstrates the good accuracy and robustness of the proposed envelope model across the compiled test specimens. In contrast, significant discrepancies are observed between experimental results and envelope models provided by ASCE 41-17 [13] and Leon [20].

Due to the limitation of experimental data, this study only assesses the applicability of the proposed envelope model to three failure modes. It is important to note that this study focuses specifically on beam-to-column T-stub connections connected to hot-rolled H-shaped steel members. For scenarios involving composite T-stub connections with concrete slabs, modifications to the developed model may be necessary to account for these variations.

CRediT authorship contribution statement

Yanfei Shen: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Mao Li:** Writing – review & editing, Investigation. **Yong Li:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Table A1
Details of the section types in the database compiled.

Ref.	T-stub	Beam	Column
[36]	Hot-rolled WT	Hot-rolled W	Hot-rolled W
[21]	Cut from hot-rolled W	Hot-rolled W	Hot-rolled W
[37–38]	Cut from hot-rolled W	Hot-rolled W	Hot-rolled W
[24]	Hot-rolled T	Hot-rolled H	Hot-rolled H
[9]	Cut from hot-rolled H	Hot-rolled H	Hot-rolled H
[39]	Cut from hot-rolled H	Hot-rolled H	Hot-rolled H
[40]	10 cut from hot-rolled H, while 2 built up from welding	Hot-rolled H	Hot-rolled H
[41]	Cut from hot-rolled H	Hot-rolled H	Hot-rolled H
[42–43]	Built up from welding	Hot-rolled W	Hot-rolled W
[35]	Cut from hot-rolled H	Hot-rolled H	Hot-rolled H
[44]	Cut from hot-rolled H	Hot-rolled H	Hot-rolled H

Table A2
Details of the bolts and T-stubs of the collected specimens subjected to cyclic loading.

Ref.	Specimen	Bolt						T-stub		
		Grade	No. T	No. S	d_b (mm)	d_{sb}	f_{ub} (MPa)	$D_T \times W_T \times t_{Ts} \times t_{Tf}$ (mm)	f_y (MPa)	f_u
[36]	B4BC1-E	A307	8	8	19.1	19.1	607	$343 \times 254 \times 13 \times 19$	350	527
	B4BC3-E	A325	8	8	19.1	19.1	862	$343 \times 254 \times 13 \times 19$	350	527
[21]	FS-03	A490	8	8	22.2	22.2	1034	$318 \times 178 \times 10 \times 14$	417	565
	FS-04	A490	8	8	25.4	25.4	1034	$356 \times 178 \times 10 \times 14$	417	565
	FS-05	A490	8	10	22.2	22.2	1034	$384 \times 264 \times 14 \times 25$	359	496
	FS-06	A490	8	8	25.4	25.4	1034	$356 \times 264 \times 14 \times 25$	359	496
	FS-07	A490	8	10	22.2	22.2	1034	$384 \times 213 \times 14 \times 24$	369	500
	FS-08	A490	8	10	25.4	25.4	1034	$432 \times 213 \times 14 \times 24$	371	500
[37–38]	Specimen 1	A490	8	4	31.8	32.0	1010	$730 \times 302 \times 24 \times 44$	441	545
	Specimen 2	A490	8	2	31.8	32.0	1010	$730 \times 302 \times 24 \times 45$	441	545
[24]	T-1	10.9	4	4	20.0	20.0	1000	$150 \times 300 \times 10 \times 15$	296	431
	T-2	10.9	4	4	20.0	20.0	1000	$250 \times 300 \times 10 \times 15$	296	431
[9]	TS-CYC 04	10.9	4	8	20.0	20.0	1000	$318 \times 257 \times 10 \times 25$	295	520
[39]	T-0	10.9	4	4	20.0	20.0	1040	$200 \times 200 \times 10 \times 16$	290	420
[40]	T-1	10.9	4	4	20.0	20.0	1040	$200 \times 200 \times 8 \times 12$	315	440
	Base	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	263	441
	T1	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 12 \times 19$	263	441
	T2	10.9	4	4	20.0	20.0	1108	$250 \times 280 \times 12 \times 19$	263	441
[41]	C1	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	263	441
	BH	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	263	441
	B9	10.9	4	4	24.0	24.0	1108	$250 \times 200 \times 10 \times 15$	263	441
	Base	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	385	450
	LL	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	385	450
	TL	10.9	4	6	20.0	20.0	1108	$250 \times 200 \times 12 \times 19$	385	450
	TJ	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	385	450
	TJH	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 12 \times 19$	385	450
	TJL	10.9	4	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	385	450
	TJS	10.9	8	4	20.0	20.0	1108	$250 \times 200 \times 10 \times 15$	385	450
[42–43]	SS-0-E	A490	8	8	35.0	34.9	1246	$485 \times 340 \times 22 \times 25$	293	445
	SS-0-W	A490	8	8	35.0	34.9	1246	$485 \times 340 \times 22 \times 25$	293	445
	SS-02a	A490	8	8	35.0	34.9	1246	$510 \times 340 \times 22 \times 50$	293	445
	SS-02b	A490	8	8	35.0	34.9	1246	$510 \times 340 \times 22 \times 50$	293	445
[35]	PJD1-N	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 8 \times 12$	260	456
	PJD1-S	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 8 \times 12$	260	456
	PJD2-N	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 9 \times 14$	260	456
	PJD2-S	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 9 \times 14$	260	456
	KJD1-N	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 8 \times 12$	260	456
	KJD1-S	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 8 \times 12$	260	456
	KJD2-N	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 9 \times 14$	260	456
	KJD2-S	10.9	4	4	22.0	22.0	1040	$270 \times 200 \times 9 \times 14$	260	456
[44]	JATD12-N	10.9	4	4	22.0	22.0	1000	$270 \times 200 \times 8 \times 12$	257	409
	JATD12-S	10.9	4	4	22.0	22.0	1000	$270 \times 200 \times 8 \times 12$	257	409
	JATD14-N	10.9	4	4	22.0	22.0	1000	$270 \times 200 \times 9 \times 14$	257	409
	JATD14-S	10.9	4	4	22.0	22.0	1000	$270 \times 200 \times 9 \times 14$	257	409

Table A3

Details of the connected beams and columns, and experimental results of the collected specimens subjected to cyclic loading.

Ref.	Specimen	Beam			Column			$M_{max-Exp}$ or $M_{max-Exp}^*$	θ_{c-Exp} or θ_{c-Exp}^*	Failure mode
		$D_b \times W_b \times t_{bw} \times t_{bf}$ (mm)	f_y (MPa)	f_u	$D_c \times W_c \times t_{cw} \times t_{cf}$ (mm)	f_y (MPa)	f_u			
								kN•m	(rad)	
[36]	B4BC1-E	457 × 191 × 10 × 14	321	487	359 × 371 × 13 × 19	307	451	284.4	0.0158	NP
	B4BC3-E	457 × 191 × 10 × 14	321	487	359 × 371 × 13 × 19	307	451	411.4	0.0176	NP
[21]	FS-03	524 × 165 × 10 × 11	368	482	375 × 394 × 17 × 27	386	510	668.0	0.0243	FM4
	FS-04	524 × 165 × 10 × 11	368	482	375 × 394 × 17 × 27	386	510	681.8	0.0267	FM4
	FS-05	600 × 178 × 10 × 13	423	528	375 × 394 × 17 × 27	386	510	1043.5	0.0298	FM6
	FS-06	600 × 178 × 10 × 13	423	528	375 × 394 × 17 × 27	386	510	1008.9	0.0258	FM6
	FS-07	600 × 178 × 10 × 13	423	528	375 × 394 × 17 × 27	386	510	1077.5	0.0243	FM6*
	FS-08	600 × 178 × 10 × 13	376	473	375 × 394 × 17 × 27	386	510	963.6	0.0207	FM6
[37-38]	Specimen 1	911 × 305 × 16 × 24	390	513	416 × 406 × 30 × 48	359	455	5188.4	0.0100	NP
	Specimen 2	911 × 305 × 16 × 24	390	513	416 × 406 × 30 × 48	359	455	4932.9	0.0067	FM4*
[24]	T-1	300 × 200 × 8 × 12	296	431	200 × 200 × 12 × 12	296	431	90.7	NA	NP
	T-2	300 × 200 × 8 × 12	296	431	200 × 200 × 12 × 12	296	431	96.3	NA	NP
[9]	TS-CYC 04	269 × 132 × 7 × 10	396	540	203 × 200 × 9 × 15	406	522	191.9	NA	FM1
[39]	T-0	300 × 200 × 8 × 12	290	420	200 × 200 × 12 × 12	290	420	144.4	NA	NP
	T-1	300 × 200 × 8 × 12	315	440	200 × 200 × 12 × 12	315	440	132.9	NA	NP
[40]	Base	294 × 200 × 8 × 12	263	441	300 × 300 × 10 × 15	263	441	193.3	0.0408	FM1
	T1	294 × 200 × 8 × 12	263	441	300 × 300 × 10 × 15	263	441	234.9	0.0482	FM6
	T2	294 × 200 × 8 × 12	263	441	300 × 300 × 10 × 15	263	441	257.4	0.0584	FM6
	C1	294 × 200 × 8 × 12	263	441	350 × 350 × 12 × 19	263	441	203.7	0.0379	FM1
	BH	250 × 200 × 9 × 14	263	441	300 × 300 × 10 × 15	263	441	171.6	0.0453	FM1
	B9	294 × 200 × 8 × 12	263	441	300 × 300 × 10 × 15	263	441	205.3	0.0405	FM1
[41]	Base	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	193.9	0.0397	FM1
	LL	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	151.4	0.0467	FM1
	TL	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	152.3	0.0321	FM6
	TJ	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	181.7	0.0317	FM1
	TJH	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	169.5	0.0338	NP
	TJL	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	181.0	0.0350	FM6
	TJS	294 × 200 × 8 × 12	385	450	300 × 300 × 10 × 15	275	433	201.5	0.0359	FM1
[42-43]	SS-0-E	600 × 250 × 12 × 16	293	445	800 × 450 × 35 × 63	293	445	897.4	NA	NP
	SS-0-W	600 × 250 × 12 × 16	293	445	800 × 450 × 35 × 63	293	445	900.7	0.0233	NP
	SS-02a	600 × 250 × 12 × 16	293	445	800 × 450 × 35 × 63	293	445	1221.0	0.0293	FM6
	SS-02b	600 × 250 × 12 × 16	293	445	800 × 450 × 35 × 63	293	445	1219.4	0.0225	FM6*
[35]	PJD1-N	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	133.0	0.1040	NP
	PJD1-S	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	120.6	0.0910	FM1
	PJD2-N	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	153.4	0.1110	NP
	PJD2-S	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	156.0	0.1240	NP
	KJD1-N	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	83.1	0.0770	FM1
	KJD1-S	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	83.5	0.0690	FM1
	KJD2-N	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	110.6	0.0930	NP
	KJD2-S	350 × 175 × 7 × 11	270	460	300 × 300 × 10 × 15	271	457	108.7	0.0855	NP
[44]	JATD12-N	350 × 175 × 7 × 11	276	436	300 × 300 × 10 × 15	293	450	96.6	0.1217	FM1
	JATD12-S	350 × 175 × 7 × 11	276	436	300 × 300 × 10 × 15	293	450	106.4	0.0845	FM1
	JATD14-N	350 × 175 × 7 × 11	276	436	300 × 300 × 10 × 15	293	450	127.4	0.0767	FM1
	JATD14-S	350 × 175 × 7 × 11	276	436	300 × 300 × 10 × 15	293	450	98.6	0.0435	NP

Note: No.T: number of tension bolts on one T-flange; No.S: number of shear bolts on one T-stem; NA: not available; NP: not experience failure; FM6*: FM6 accompanied by beam flange fracturing along the last bolt-row; FM4*: FM4 for specimen whose T-stem was welded to the beam flange; $M_{max-Exp}^*$: measured maximum moment for specimens that didn't experience failure; $\theta_{max-Exp}^*$: rotation corresponding to $M_{max-Exp}^*$.

Appendix B

The procedure for developing the expressions for T_1' , T_2' and T_3'

(1) FM1

Force equilibrium of the half T-flange for FM1 depicted in Fig.7 results in

$$M_p + Qn - \frac{T_1'}{2}m = 0 \quad (A1)$$

and

$$\delta M_p = Qn \quad (A2)$$

T_1' is then determined by

$$T_1' = \frac{2(1 + \delta)M_p}{m} \quad (A3)$$

M_p is given by

$$M_p = \left(\frac{wt_{fT}^2}{4} \right) (c_{\beta} f_y) \quad (A4)$$

Submitting Eq. (A4) into Eq. (A3), gives

$$T'_1 = \frac{(1 + \delta)w_{TF}^2 c_{\beta} f_y}{2m} \quad (A5)$$

(2) FM2

Force equilibrium of the half T-flange shown for FM2 depicted in Fig. 7 results in

$$M_p + B_u n - \frac{T'_1}{2}(m + n) = 0 \quad (A6)$$

Submitting Eq. (A4) into Eq. (A6), gives

$$T'_2 = \frac{\frac{1}{2}w_{TF}^2 c_{\beta} f_y + 2nB_u}{m + n} \quad (A7)$$

where c_{β} is taken as unity for FM2.

(3) FM3

Force equilibrium of the T-flange for FM3 depicted in Fig.7 results in

$$T'_3 = 2B_u \quad (A8)$$

References

- A.S. Elnashai, F.A. Danesh-Ashtiani, A.Y. Elghazouli, Response of semi-rigid steel frames to cyclic and earthquake loads, *J. Struct. Eng.* 124 (8) (1998) 857–867, [https://doi.org/10.1061/\(ASCE\)0733-9445\(1998\)124:8\(857\)](https://doi.org/10.1061/(ASCE)0733-9445(1998)124:8(857)).
- M.E. Lemonis, Steel moment resisting frames with both joint and beam dissipation zones, *J. Constr. Steel Res.* 147 (2018) 224–235, <https://doi.org/10.1016/j.jcsr.2018.03.020>.
- M.E. Lemonis, G.D. Hatzigeorgiou, P.G. Asteris, Seismic behaviour of irregular steel frames with beam and joint energy dissipation, *Soil Dyn. Earthquake Eng.* 152 (2022) 107052, <https://doi.org/10.1016/j.soildyn.2021.107052>.
- G.Q. Li, W.L. Shi, J.F. Wang, *Design of Steel Frames With Semi-Rigid Connections*, China Architecture & Building Press, Beijing, China, 2009.
- M. Wang, Y.S. Tong, Y. Sun, L. Yang, Structural fuse design of bolted steel frame with low yield point angle steel connection components, *Thin-Wall. Struct.* 179 (2022) 109553, <https://doi.org/10.1016/j.tws.2022.109553>.
- H.T. Thai, S.E. Kim, Second-order distributed plasticity analysis of steel frames with semi-rigid connections, *Thin-Wall. Struct.* 94 (2015) 120–128, <https://doi.org/10.1016/j.tws.2015.04.011>.
- V. Saberi, M. Gerami, A. Kheyroddin, Comparison of bolted end plate and T-stub connection sensitivity to component thickness, *J. Constr. Steel Res.* 98 (2014) 134–145, <https://doi.org/10.1016/j.jcsr.2014.02.012>.
- H.H. Ma, A.M. Issa, F. Fan, G.O. Adeoti, An experimental and numerical study of a semi-rigid bolted-plate connections (BPC), *Thin-Wall. Struct.* 88 (2015) 82–89, <https://doi.org/10.1016/j.tws.2014.11.011>.
- F. Iannone, M. Latour, V. Piluso, V.G. Rizzano, Experimental analysis of bolted steel beam-to-column connections: component identification, *J. Earthquake Eng.* 15 (2) (2011) 214–244, <https://doi.org/10.1080/13632461003695353>.
- FEMA 445, *Next-Generation Performance Based Seismic Design Guidelines*, Program Plan For New and Existing Buildings, Applied Technology Council for the Federal Emergency Management Agency, Washington, D.C., 2018 prepared by.
- Guidelines for Performance-Based Design of Tall Buildings, PEER Report, Tall Buildings Initiative Working Group, prepared by Pacific Earthquake Engineering Research Center, Berkeley, California, 2017.
- D.G. Lignos, A.R. Hartloper, A. Elkady, G.G. Deierlein, R. Hamburger, Proposed updates to the ASCE 41 nonlinear modeling parameters for wide-flange steel columns in support of performance-based seismic engineering, *J. Struct. Eng.* 145 (9) (2019) 04019083, [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002353](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002353).
- ASCE 41-17, *Seismic Evaluation and Retrofit of Existing Buildings*, American Society of Civil Engineers, Reston, 2017.
- FEMA P-58-1, *Seismic Performance Assessment of Buildings-Volume1 Methodology*, 2nd edition, Applied Technology Council for the Federal Emergency Management Agency, Washington, D.C., 2018. ; prepared by.
- Guidelines For Nonlinear Structural Analysis For Design of buildings: Part IIa—steel moment Frames, prepared by Applied Technology Council for National Institute of Standards and Technology, Gaithersburg, Maryland, 2017.
- V. Sharma, M.K. Shirmali, S.D. Bharti, T.K. Datta, Seismic fragility evaluation of semi-rigid frames subjected to near-field earthquakes, *J. Constr. Steel Res.* 176 (2021) 106384, <https://doi.org/10.1016/j.jcsr.2020.106384>.
- R.T. Leon, D.H. Kim, Seismic performance of older PR frames in areas of infrequent seismicity, in: *13th World Conference on Earthquake Engineering*, Vancouver, BC, 2004.
- D.H. Kim, R.T. Leon, Fragility analyses of mid-rise T-stub PR frames in the Mid-America earthquake region, *Int. J. Steel Struct.* 13 (2013) 81–91, <https://doi.org/10.1007/s13296-013-1008-7>.
- FEMA 356, *Prestandard and Commentary for the Seismic Rehabilitation of Buildings*, American Society of Civil Engineers for Federal Emergency Management Agency, Washington, D.C., 2000. Prepared by.
- R.T. Leon, Tests On T-Stub Connections - SAC Phase Subtask 7.03, Georgia Institute of Technology, Atlanta, USA, 1999.
- J.M. Smallidge, Master's thesis, Georgia Institute of Technology, Atlanta (GA), 1999.
- J.A. Swanson, Ph.D. Thesis, Georgia Institute of Technology, Atlanta, GA, 1999.
- EN 1993-1-8:2005 (E), *Design of Steel Structures - Part 1-8: Design of Joints*, European Committee for Standardization, Brussels, Belgium, 2005.
- Q.C. Duan, Master's thesis, Wuhan University of Technology, China, 2003.
- J.P. Jaspart, R. Maquoi, Prediction of the semi-rigid and partial strength properties of structural joints, in: *SSRC Annual technical Session and Meeting*, Lehigh, 1994.
- J.P. Jaspart, K. Weynand, M. Steenhuis, The stiffness model of revised Annex J of Eurocode 3, in: *Proceedings of the Third International Workshop on Connections in Steel Structures*, Trento, Italy, 1995.
- Y. Wang, *Semi-rigid Steel connections: Design Theory and Engineering Application*, 1st edition, China Architecture & Building Press, Beijing, 2011.
- D.A. Nethercot, Steel beam to column connections—A review of test data and their applicability to the evaluation of the joint behaviour of the performance of steel frames. Rep. No. RR1084, Sheffield, UK, 1984.
- W.F. Chen, N. Kishi, Semi rigid steel beam to column connections: database and modeling, *J. Struct. Eng.* 115 (1) (1989) 105–119, [https://doi.org/10.1061/\(ASCE\)0733-9445\(1989\)115:1\(105\)](https://doi.org/10.1061/(ASCE)0733-9445(1989)115:1(105)).
- K.M. Abdalla, W.F. Chen, Expanded database of semi-rigid steel connections, *Comput. Struct.* 56 (4) (1995) 553–564, [https://doi.org/10.1016/0045-7949\(94\)00558-K](https://doi.org/10.1016/0045-7949(94)00558-K).
- K. Weynand, M. Huter, P. Kirby, L.A.P. Simões da Silva, P.J.S. Cruz, SERICON – a databank for tests on semi-rigid joints, in: *COST C1 International Conference on Control of the Semi-Rigid Behaviour of Civil Engineering Structural Connections*, Liège, Belgium, 1998.
- L. Mak, A. Elkady, Experimental database for steel flush end-plate connections, *J. Struct. Eng.* 147 (7) (2021) 04721006, [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0003064](https://doi.org/10.1061/(ASCE)ST.1943-541X.0003064).
- B. David, Z. Lynn, SAC Connection Database, SAC Steel Project, Richmond, CA, 2000. <https://www.sacsteel.org/connections/index.html>.
- D.G. Lignos, H. Krawinkler, A database in support of modeling of component deterioration for collapse prediction of steel frame structures, in: *Proc. ASCE Structures Congress*, SEI institute, Long Beach, CA, 2007.
- X. Bu, Q. Gu, X.W. Wang, Experimental study on the seismic performance of three-dimensional beam to middle column connection with t-stub, *J. Eng. Mech.* 34 (8) (2017) 105–116, <https://doi.org/10.6052/j.issn.1000-4750.2016.03.0176>.
- G.P. Forcier, Ph.D. thesis, University of Minnesota, 1994.
- E.P. Popov, S.M. Takhirov, Bolted large seismic steel beam-to-column connections Part 1: experimental study, *Eng. Struct.* 24 (12) (2002) 1523–1534, [https://doi.org/10.1016/S0141-0296\(02\)00086-X](https://doi.org/10.1016/S0141-0296(02)00086-X).
- E.P. Popov, S.M. Takhirov, Experimental Study of Large Seismic Steel Beam-to-Column Connections, PEER Report, Pacific Earthquake Engineering Research Center, University of California, Berkeley, 2000.
- H.Y. Peng, Master's thesis, Changsha University of Science & Technology, China, 2011.
- X.W. Zheng, Master's thesis, Guangxi University, China, 2013.
- X.H. Li, Master's thesis, Guangxi University, China, 2016.
- R. Herrera, C. Salas, J.F. Beltran, E. Nunez, Experimental performance of double built-up T moment connections under cyclic loading, *J. Constr. Steel Res.* 138 (2017) 742–749, <https://doi.org/10.1016/j.jcsr.2017.08.022>.
- Q. Constanza Salas, Ph.D. Thesis, University of Chile, Santiago, Chile, 2016.

- [44] Q. Shi, S.L. Yan, L.L. Kong, X. Bu, X.W. Wang, H.S. Sun, Seismic behavior of semi-rigid steel joints- Major axis T-stub and minor axis end-plate, *J. Constr. Steel Res.* 159 (2019) 476–492, <https://doi.org/10.1016/j.jcsr.2019.04.036>.
- [45] *Steel Construction manual*[M], American Institute of Steel Construction, Chicago, IL, 2005.
- [46] *GB 50017-2017, Standard For Design of Steel Structures*, China Architecture and Building Press, Beijing, China, 2017.
- [47] *AISC 360-22, Specification For Structural Steel Buildings*, American Institute of Steel Construction, Chicago, IL, 2022.
- [48] *Specification For Structural Joints Using High-Strength Bolts*, RCSC Committee A.1, Research Council on Structural Connections, Chicago, 2020. Prepared by.
- [49] A. Elkady, Response characteristics of flush end-plate connections, *Eng. Struct.* 269 (2022) 114856, <https://doi.org/10.1016/j.engstruct.2022.114856>.